



Increasing the Trigger and Selection Efficiency for Higgs to Diphoton Signal at the CMS Experiment Using Machine Learning Techniques

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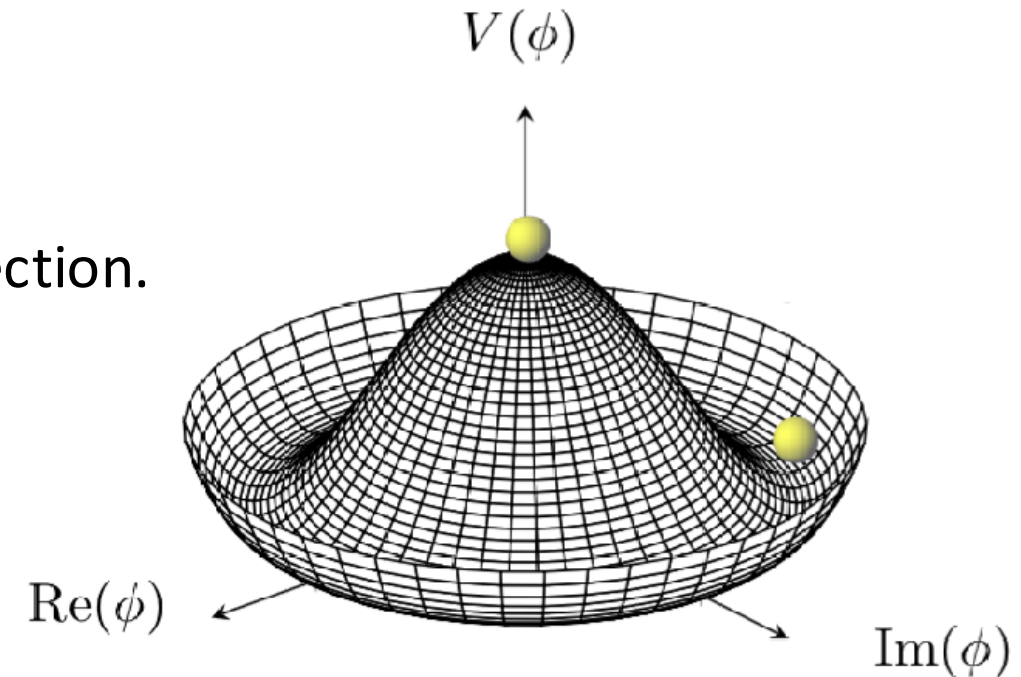
- The Standard Model and Higgs Boson
- CMS Detector, Triggering, and Analysis
- Boosted Decision Tree (BDT) Basics
- Development of BDT-Based High-Level Trigger (MVA-HLT)
- Development of BDT-Based Offline Selection
- Improvements to Signal Efficiency for Selected Samples
 - Trigger-level efficiency
 - Offline selection efficiency

- | | mass → | charge → | spin → |
|--------------|----------------------------------|----------|--------|
| QUARKS | $\approx 2.3 \text{ MeV}/c^2$ | $2/3$ | $1/2$ |
| | $\approx 1.275 \text{ GeV}/c^2$ | $2/3$ | $1/2$ |
| | $\approx 173.07 \text{ GeV}/c^2$ | $2/3$ | $1/2$ |
| | 0 | 0 | 1 |
| | $\approx 126 \text{ GeV}/c^2$ | 0 | 0 |
| | 0 | 0 | 1 |
| LEPTONS | $\approx 4.8 \text{ MeV}/c^2$ | $-1/3$ | $1/2$ |
| | $\approx 95 \text{ MeV}/c^2$ | $-1/3$ | $1/2$ |
| | $\approx 4.18 \text{ GeV}/c^2$ | $-1/3$ | $1/2$ |
| | 0 | 0 | 1 |
| | 0 | 0 | 1 |
| | 0 | 0 | 1 |
| GAUGE BOSONS | $0.511 \text{ MeV}/c^2$ | -1 | $1/2$ |
| | $105.7 \text{ MeV}/c^2$ | -1 | $1/2$ |
| | $1.777 \text{ GeV}/c^2$ | -1 | $1/2$ |
| | $91.2 \text{ GeV}/c^2$ | 0 | 1 |
| | $< 2.2 \text{ eV}/c^2$ | 0 | $1/2$ |
| | $< 0.17 \text{ MeV}/c^2$ | 0 | $1/2$ |
| | $< 15.5 \text{ MeV}/c^2$ | 0 | $1/2$ |
| | $80.4 \text{ GeV}/c^2$ | ± 1 | 1 |

The Brout–Englert–Higgs (BEH) Mechanism



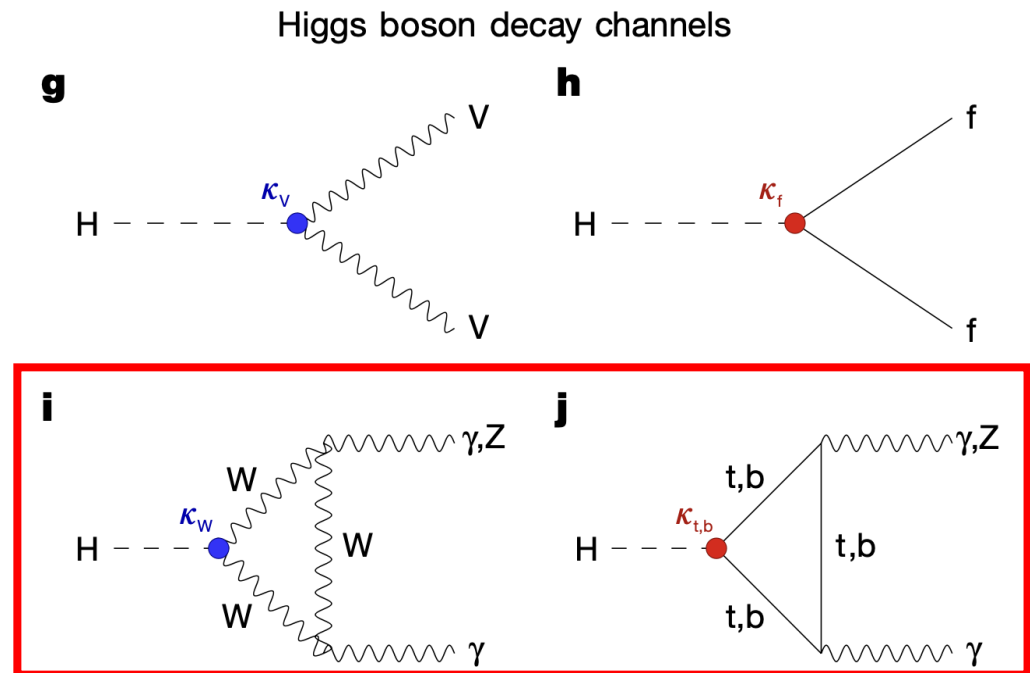
- The Higgs field was first theorized in 1964.
 - A vital change to field theory – “sombbrero” potential spontaneously breaks electroweak symmetry.
 - Also explains fermion masses through the Yukawa coupling.
 - Predicts a (scalar) Higgs boson, became major question in particle physics.
- Detection was impossible for many years.
 - Until the LHC, accelerator energy was too low.
 - Process is very rare – need high rates and good detection.



Single Higgs Boson Decay Modes



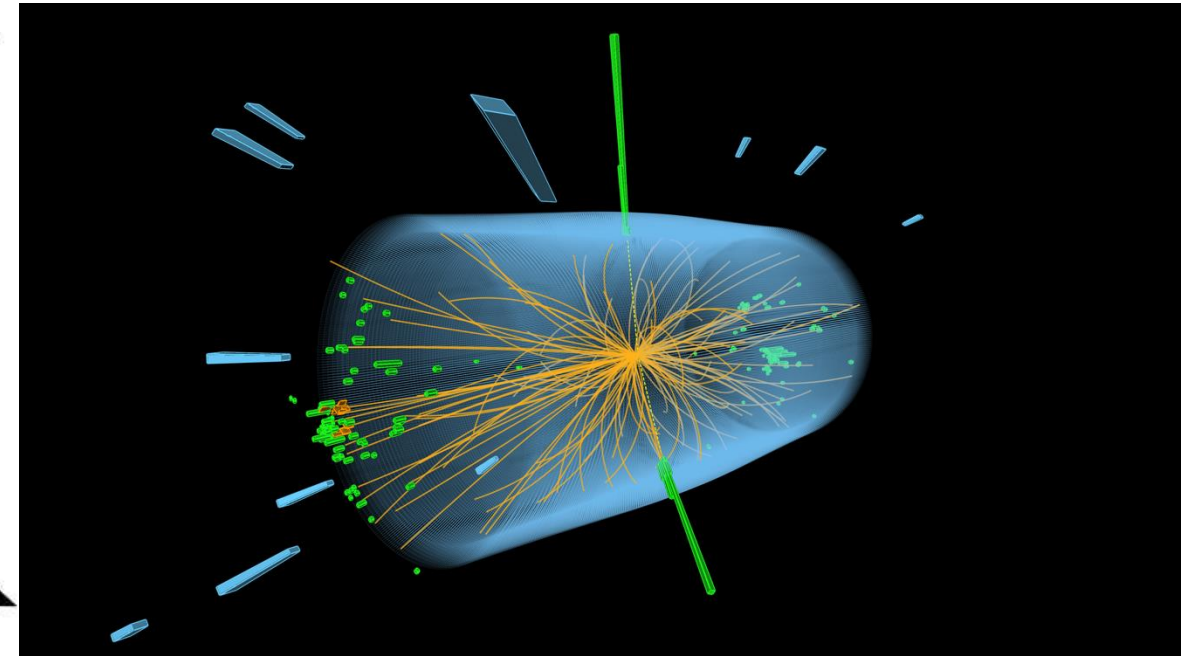
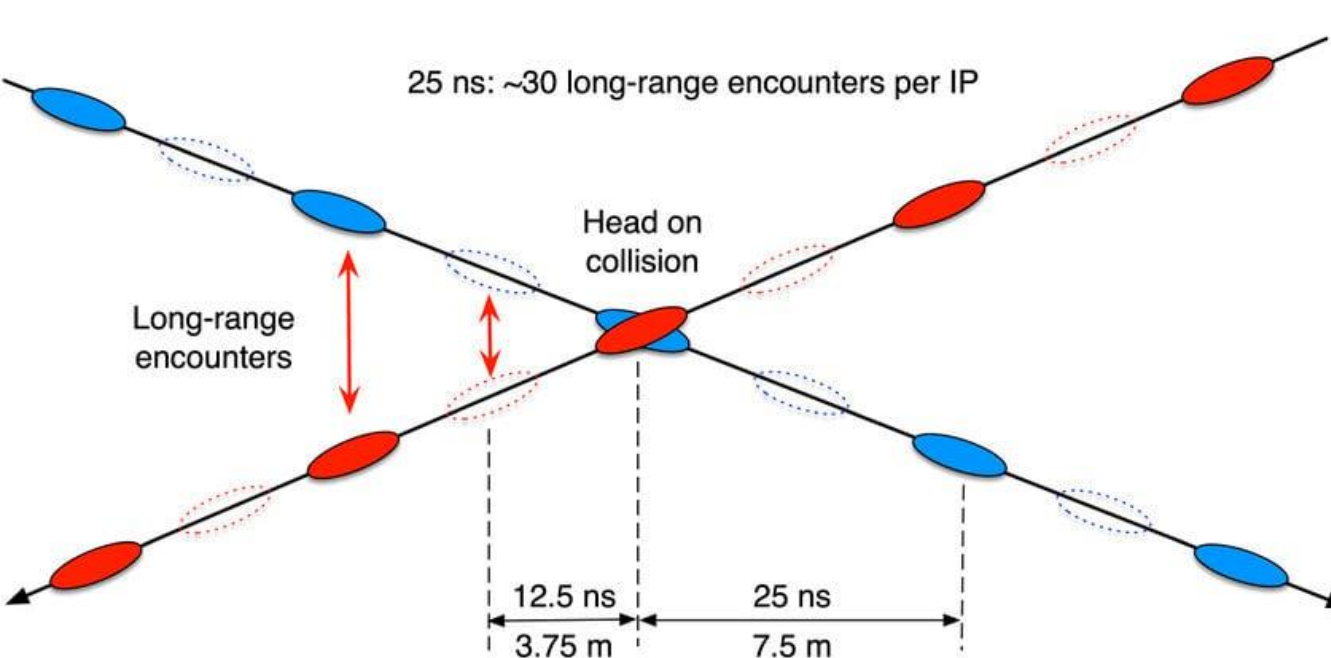
- Regardless of how it is produced, a Higgs boson will decay into one of several allowed final states.
- All decays are not equally likely – branching fractions for the Higgs into each mode can be calculated
 - **g**: WW ($\sim 22\%$) and ZZ ($\sim 2.7\%$)
 - **h**: $b\bar{b}$ ($\sim 58\%$), gg ($\sim 8\%$), $\tau\tau$ ($\sim 6\%$) ...
 - **i & j**: $\gamma\gamma$ ($\sim 0.2\%$), $Z\gamma$ ($\sim 0.2\%$)
- While $H \rightarrow \gamma\gamma$ has a relatively low BF, it provides a clean and high-resolution channel with which to measure the Higgs.



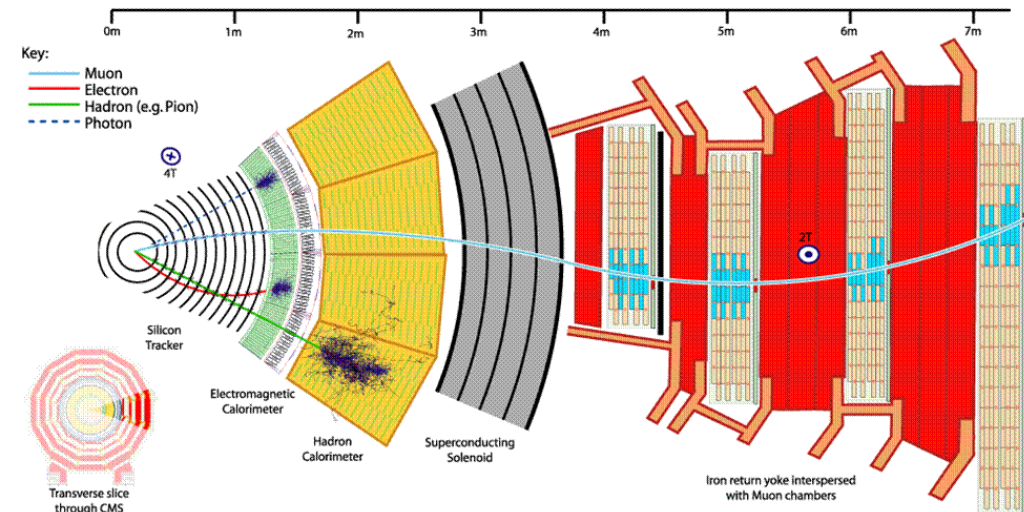
The LHC Experiment



- The LHC accelerates counterrotating “bunches” of protons to relativistic speeds and collides them at “interaction points” (IPs).
 - Each IP corresponds to a specific detector .
- Proton-proton collisions with very high energies create large quantities of outgoing particles.



- CMS has several layers of detector elements:
 - Silicon pixels and strips give spatial tracking information.
 - Calorimeters measure energy deposition in different areas.
 - Electromagnetic calorimeter (ECAL) measures electrons and photons.
 - Hadron calorimeter (HCAL) detects hadronic particles.
 - 3.8T Solenoid provides strong magnetic field for inner detector.
 - Muon chambers (Drift Tubes)
 - Scintillators and “RPCs” for triggering



CMS DETECTOR

Total weight : 14,000 tonnes
 Overall diameter : 15.0 m
 Overall length : 28.7 m
 Magnetic field : 3.8 T

STEEL RETURN YOKE
 12,500 tonnes

SILICON TRACKERS
 Pixel (100x150 μm) $\sim 1\text{m}^2 \sim 66\text{M}$ channels
 Microstrips (80x180 μm) $\sim 200\text{m}^2 \sim 9.6\text{M}$ channels

SUPERCONDUCTING SOLENOID
 Niobium titanium coil carrying $\sim 18,000\text{A}$

MUON CHAMBERS
 Barrel: 250 Drift Tube, 480 Resistive Plate Chambers
 Endcaps: 540 Cathode Strip, 576 Resistive Plate Chambers

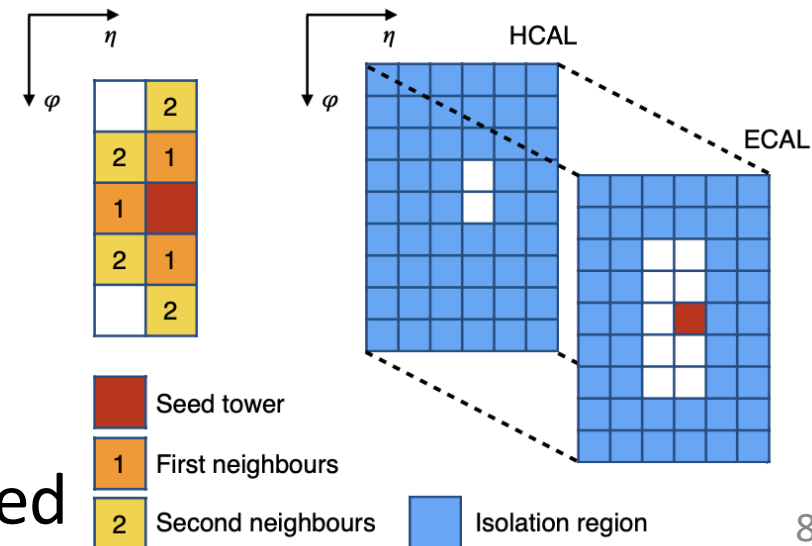
PRESHOWER
 Silicon strips $\sim 16\text{m}^2 \sim 137,000$ channels

FORWARD CALORIMETER
 Steel + Quartz fibres $\sim 2,000$ channels

CRYSTAL ELECTROMAGNETIC CALORIMETER (ECAL)
 $\sim 76,000$ scintillating PbWO_4 crystals

HADRON CALORIMETER (HCAL)
 Brass + Plastic scintillator $\sim 7,000$ channels

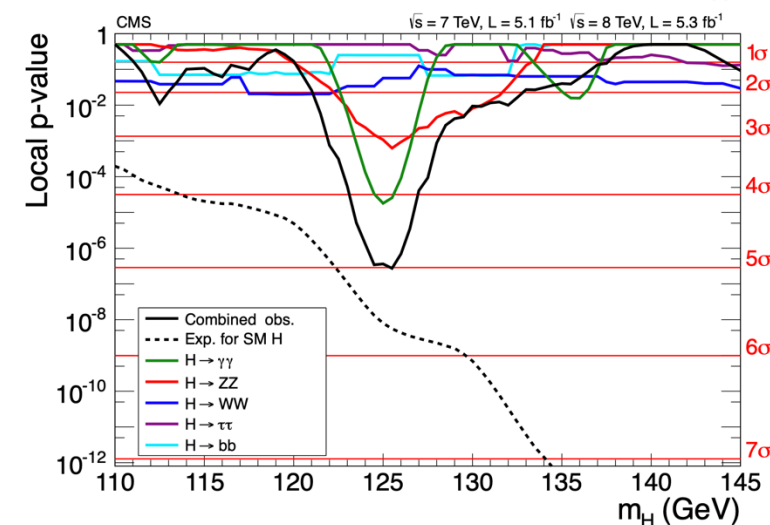
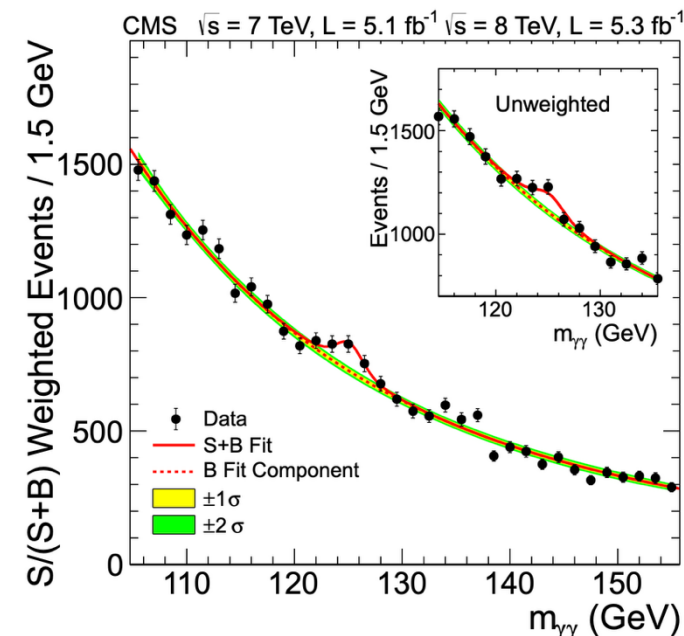
- Reconstruction of photon and electron signatures occurs almost exclusively in the Electromagnetic Calorimeter (ECAL).
- Energy deposition measured in ECAL by Lead Tungstate (PbWO_4) crystals.
 - 75,848 total crystals between one Barrel and two Endcaps.
 - Crystals arranged in azimuth(ϕ) and pseudorapidity(η).
 - Crystals are designed to be smaller than shower size, “superclustering” associates.
- “Shower shape” quantifies deposited energy shape
- “Isolation” measures energy deposited around cluster
- In conjunction with kinematics, e/γ can be reconstructed



Higgs Discovery



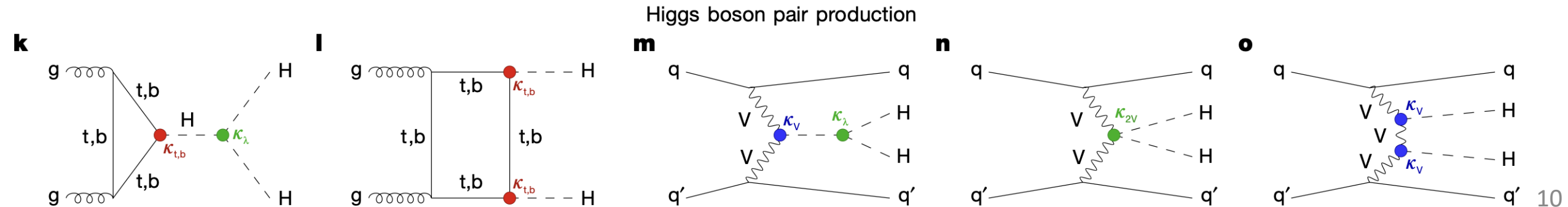
- LHC began high-energy proton-proton collisions in 2010.
 - Initial center-of-mass energy of $\sqrt{s} = 7$ TeV.
- Initial discovery occurred in 2012.
 - Found at mass of $125 \text{ GeV}/c^2$.
 - $\gamma\gamma$ and ZZ decays were vital.
- Since then, $\sqrt{s}$ and luminosity have increased.
 - Run 2 saw $\sqrt{s} = 13$ TeV.
 - Run 3(Current) brings $\sqrt{s} = 13.6$ TeV.
 - With each increase, can produce more Higgs.



DiHiggs Production Modes



- There are several modes by which two Higgs bosons can be produced.
- The coupling constants at each vertex are key parameters that must be measured to refine SM and search for Beyond Standard Model (BSM) physics.
- DiHiggs production occurs at very low rate ($\sigma_H = 54 \text{ pb}$, $\sigma_{HH} = 33 \text{ fb}$) and has not yet been detected at the LHC. DiHiggs searches are a major part of Run 3.





- Run 3 bunch crossings occur at 40 MHz (every 25 ns)
 - Each bunch crossing contains an average of 57 collisions (termed pileup)
- Two-tiered triggering system is designed to reduce rate to a level that can be read out, processed and stored
- Hardware-based Level 1 (L1) trigger takes raw detector inputs and reduces rate to ~ 110 kHz.
- Software-based High-Level Trigger (HLT) processes outputs from L1, performs event reconstruction, and selects just 7 kHz of crossings to save.
 - Each event is ~ 1 MB, so storage load is still huge

The Standard $H \rightarrow \gamma\gamma$ HLT (sHLT)



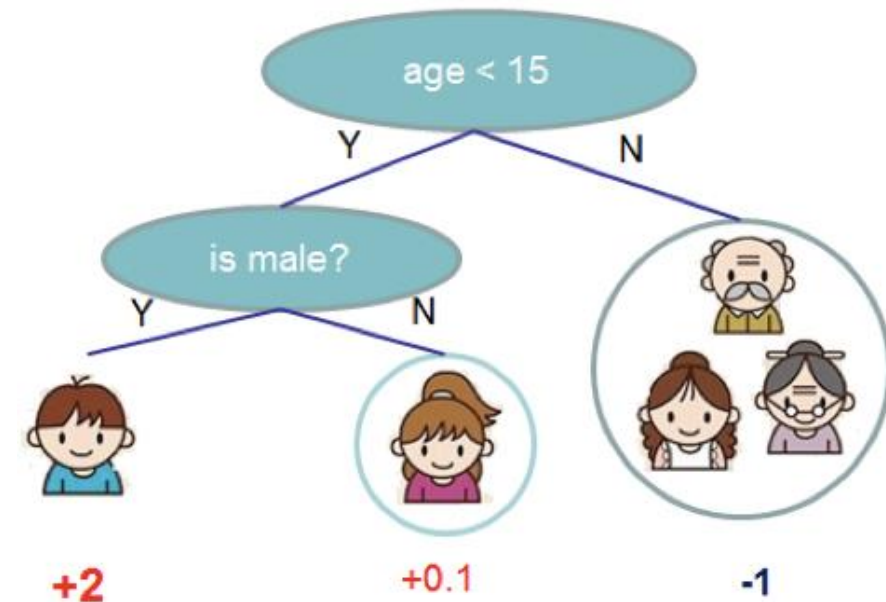
- Each analysis group develops their own specific triggers, designed to isolate the necessary components of their chosen physics event. For $H \rightarrow \gamma\gamma$, the HLT must accept two photons that are consistent with originating from the decay of a Higgs boson.
- Throughout Run 2, and much of Run 3, the primary $H \rightarrow \gamma\gamma$ HLT was based on hard cuts on kinematic, isolation, and shower shape variables.

Detector Region	H/E	R_9 (E_{3x3}/E_{SC})	ECal Isolation	Track Isolation	$\sigma_{i\eta i\eta}$	E_T (Lead , Sublead)	$M_{\gamma\gamma}$
Barrel	< 0.12	> 0.85	--	--	--	> 30, 22 GeV	> 90 GeV/c^2
	< 0.12	0.5 < R9 < 0.85 Hard cut at R9 < 0.5	< 6 GeV	< 6 GeV	< 0.015	> 30, 22 GeV	> 90 GeV/c^2
Endcap	< 0.12	> 0.9	--	--	--	> 30, 22 GeV	> 90 GeV/c^2
	< 0.12	0.8 < R9 < 0.9 Hard cut at R9 < 0.8	< 6 GeV	< 6 GeV	< 0.035	> 30, 22 GeV	> 90 GeV/c^2

Boosted Decision Trees



- Since the HLT is a binary classification problem (signal vs background), Boosted Decision Trees (BDTs) are an obvious approach.
- A BDT is a Machine Learning technique where “ensembles” of simple decision trees are combined to process high-dimensional data.
- Each tree operates on subsets of the available input variables, assigning weights at each “node” to build up a complex decision tree.
- Our BDTs will use shower shape and isolation variables.



Development of a BDT-Based High-Level Trigger (MVA-HLT)

Training Samples for a Trigger-Level Diphoton BDT



- First step – selecting signal and background samples for training.
- The goal of the $H \rightarrow \gamma\gamma$ HLT is to not only correctly identify photon candidates, but to primarily accept photon candidates originating from Higgs decay.
- The obvious signal sample is a $H \rightarrow \gamma\gamma$ Monte Carlo (MC) simulation.
 - Gluon Fusion Higgs was chosen, as it encompasses $> 85\%$ of Higgs production.
- There are many backgrounds to reject, including hadronic jets, single photon backgrounds, and real diphotons from other processes. Many of the backgrounds are very hard to simulate properly.
- Trigger-level data samples will be almost exclusively background, as Higgs production is so rare as to make up a vanishingly small proportion of the total collected data

Training a Trigger-Level (Single Photon) BDT



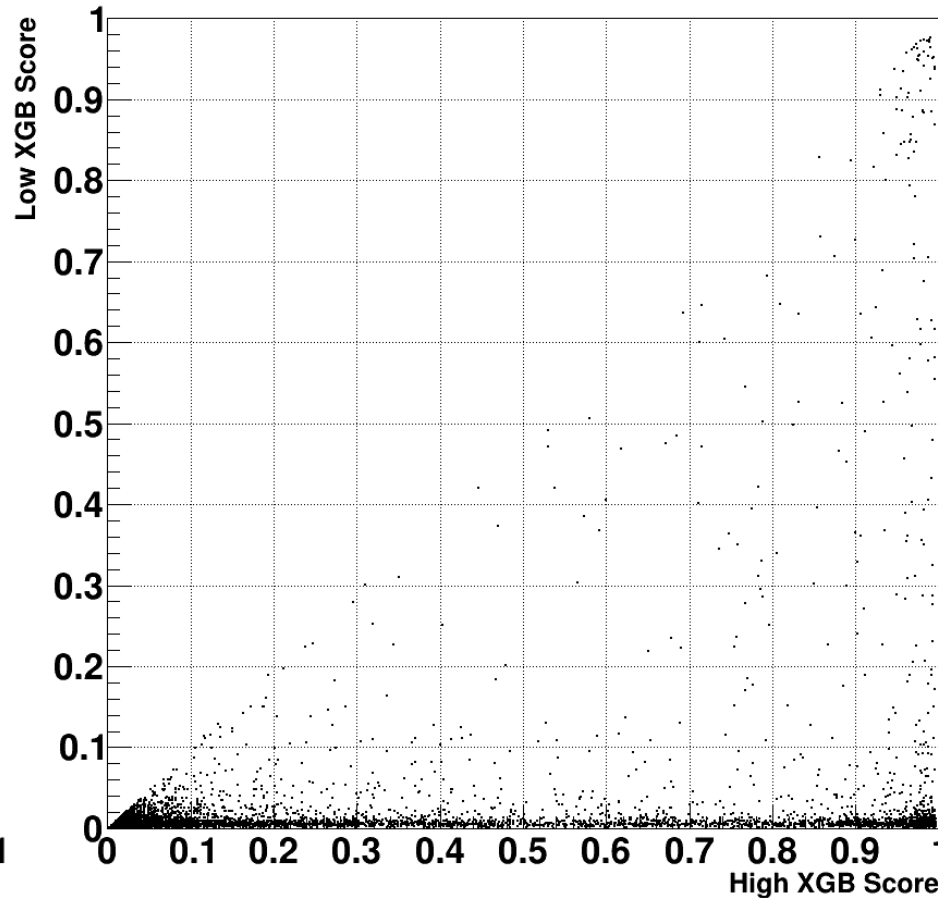
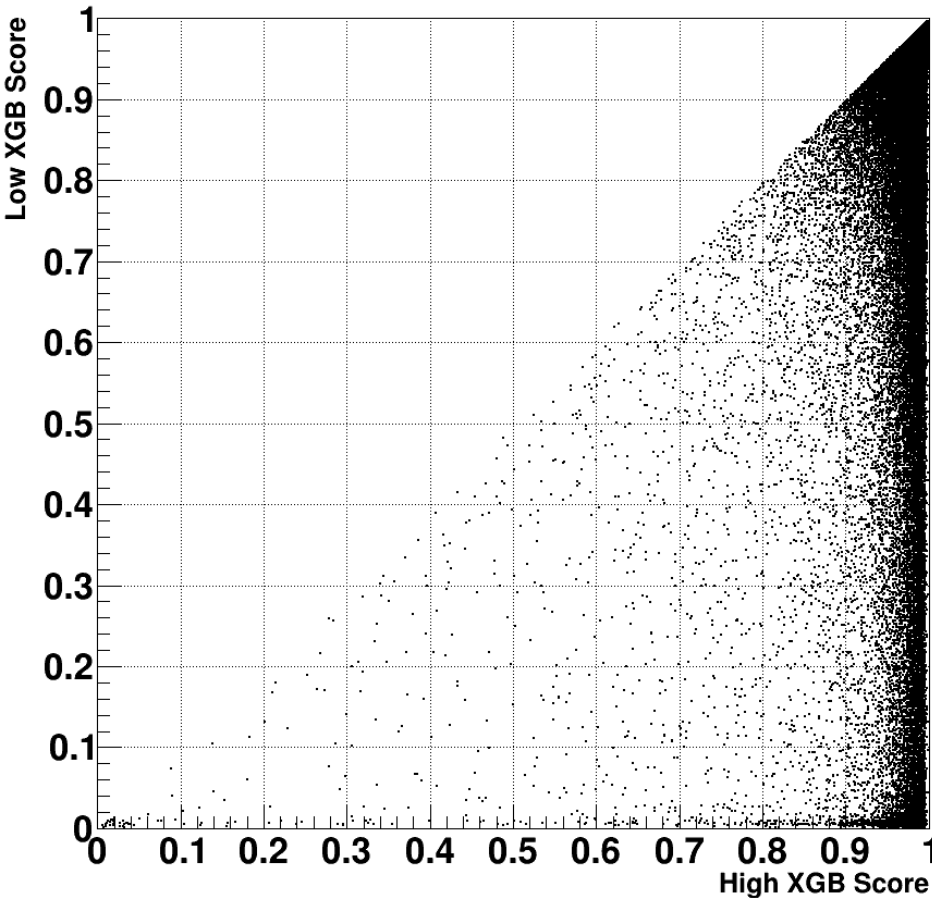
- Once the samples are selected, input variables must be chosen on which to train the BDT.
 - An obvious starting point is to use the same variables used by the standard HLT!
- Selected input variables are of three types:
 - Kinematic: Raw Supercluster energy, η_{SC}
 - Shower shape: $\sigma_{i\eta i\eta}$, R_9 , η Width, ϕ Width, H/E
 - Isolation: ECal Isolation
- Once the samples are collected and the input variables determined, the “hyperparameters” of the model must be optimized to maximize the performance of the BDT (like Max Depth, Learning Rate)
 - These hyperparameters were optimized iteratively and experimentally.

Output Trigger-Level Score Distributions



Higgs MC Signal

“Ephemeral” Data



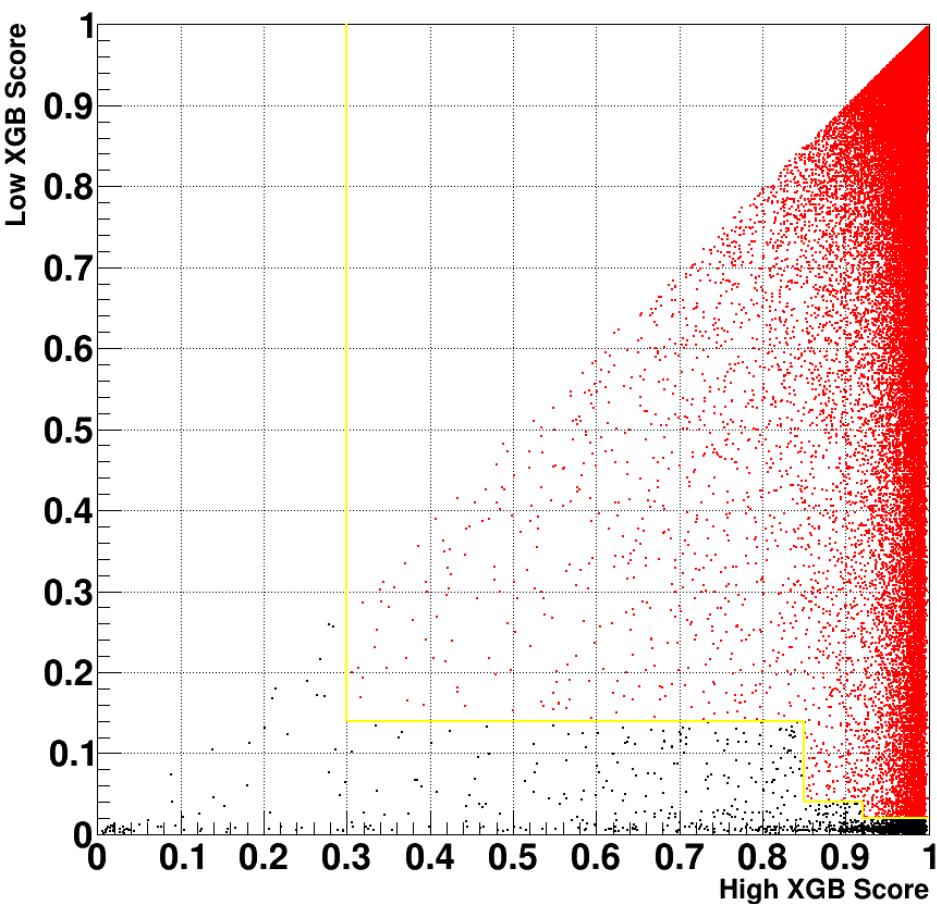
- Clearly, the signal usually has one (or both) candidate scores near 1.
- Meanwhile, most data events have one (or both) candidate scores near 0.

- “Ephemeral” data are all data passing L1 trigger, so includes all possible backgrounds.
- The outputs of this BDT are a score ranging from 0 to 1.
- A score of 0 means that the model has assigned maximum probability that the photon candidate is background, where a score of 1 means maximum probability of being signal.
- Since two photon candidates are selected for $H \rightarrow \gamma\gamma$, both scores are plotted together here.

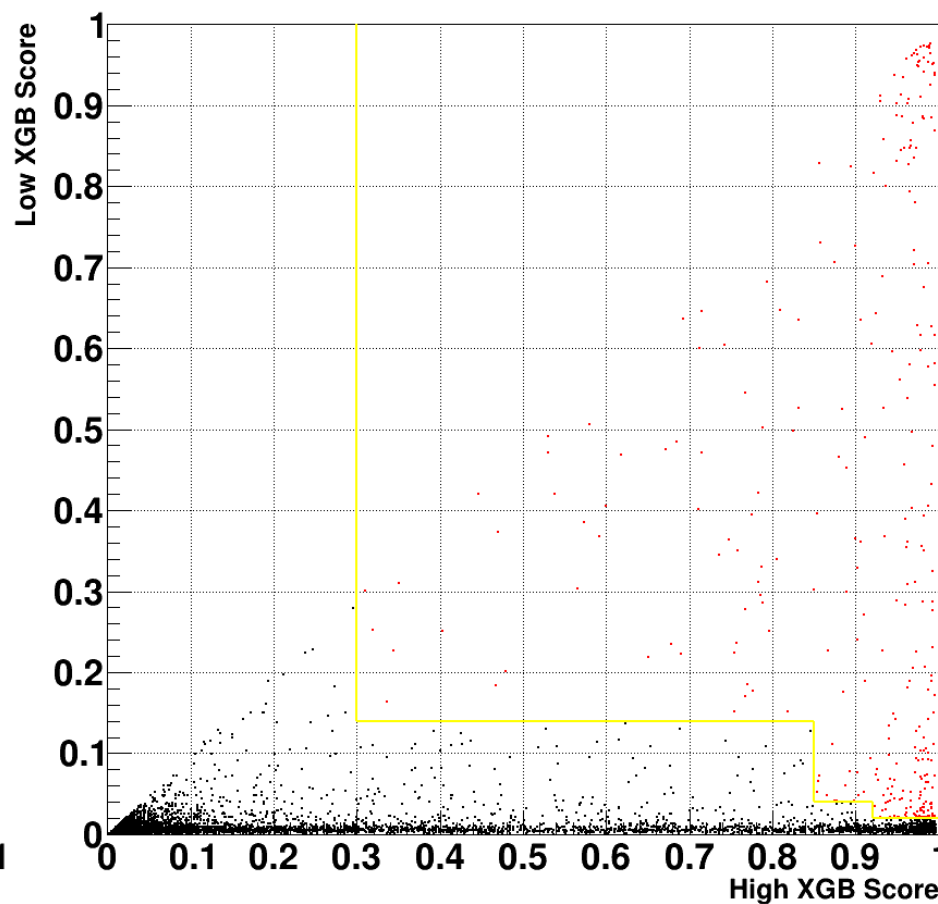
Selection of BDT Score Cuts for MVA-HLT



Higgs MC Signal



“Ephemeral” Data



- Cuts (in yellow) are chosen to select as many signal MC events as possible while keeping the data rate low.
- “Step” pattern shows 2 dimensional score cuts.
- This structure allows many events with one very high-quality photon to be preserved in the signal MC.

- Some signal losses occur due to having a second photon with a very low score.
- Very little data is accepted – This selection has a much lower rate (29.4 Hz) than the standard HLT (61.5 Hz).

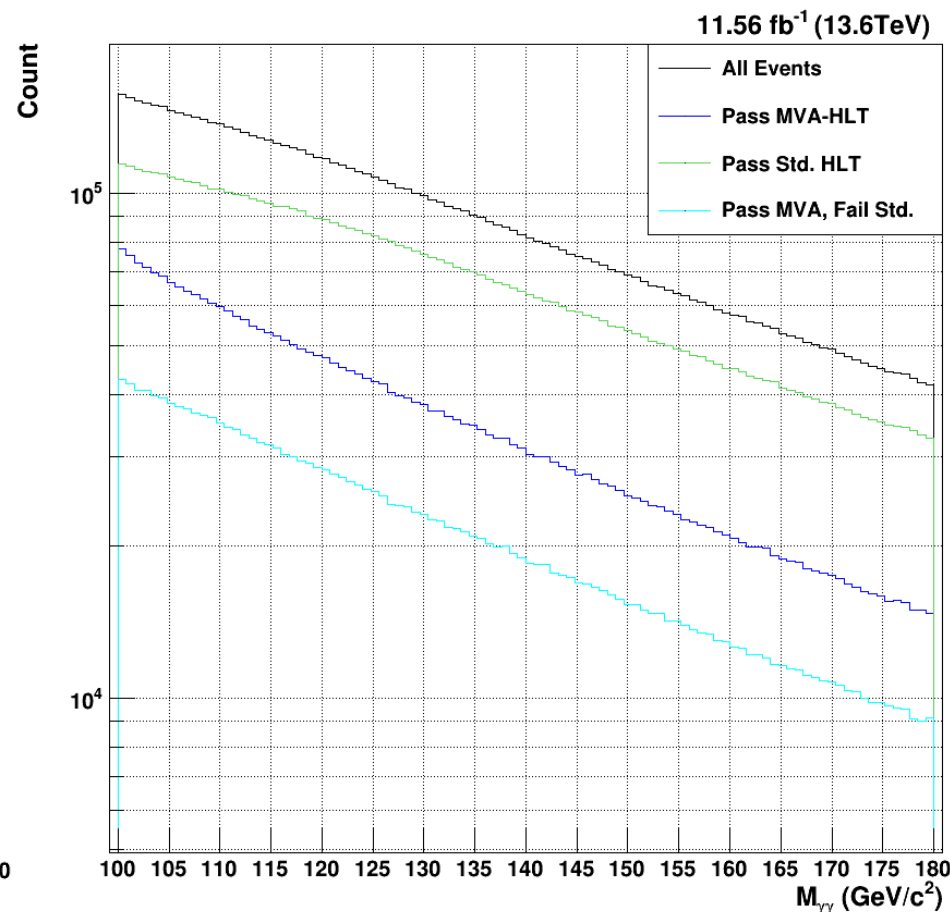
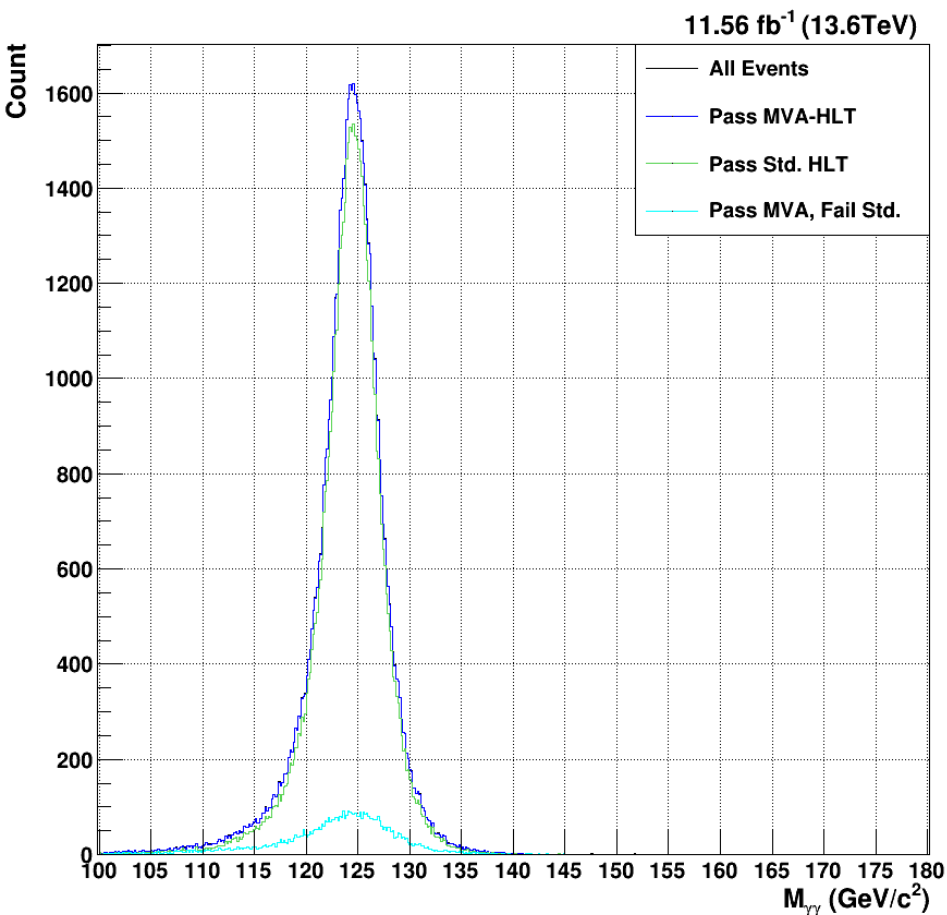
Trigger-Level Results for $H \rightarrow \gamma\gamma$ and $HH \rightarrow b\bar{b}\gamma\gamma$

Single Higgs $M_{\gamma\gamma}$



Higgs MC Signal

Data



- Much of the gain from our selection occurs within the mass peak.
- The signal efficiency of our selection (97.3%) is improved relative to the sHLT (88.7%).
- The data rate is considerably lower for our selection (~47% relative reduction in rate).

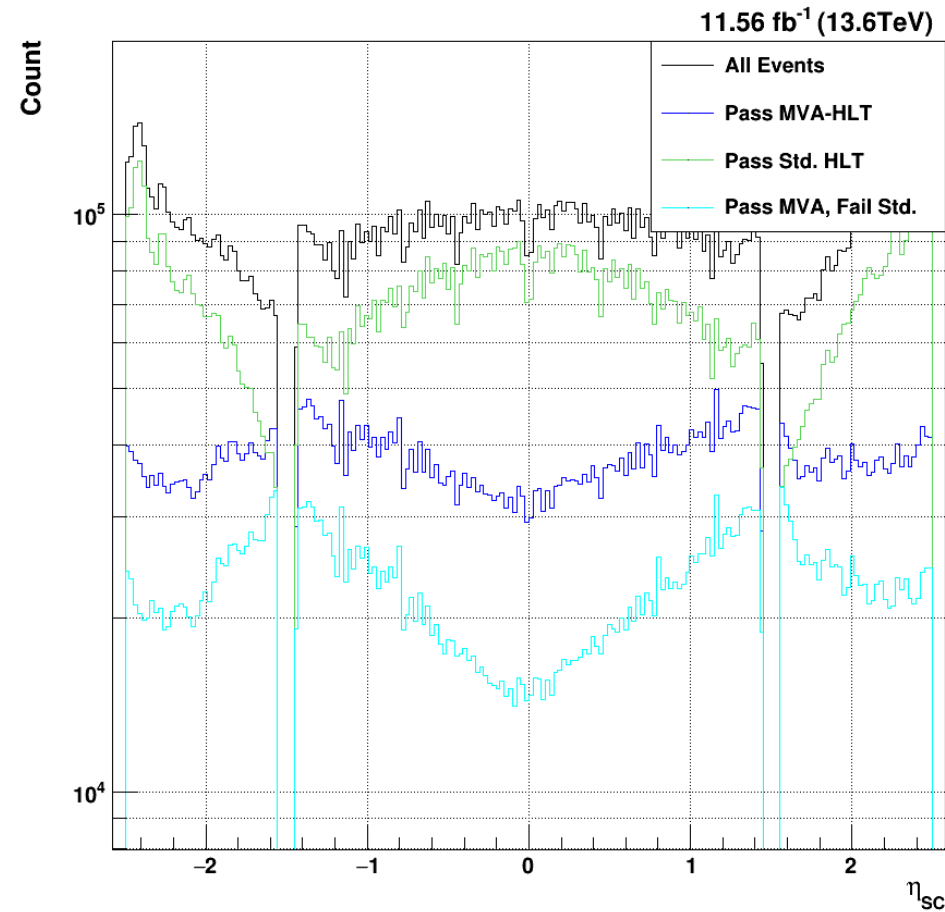
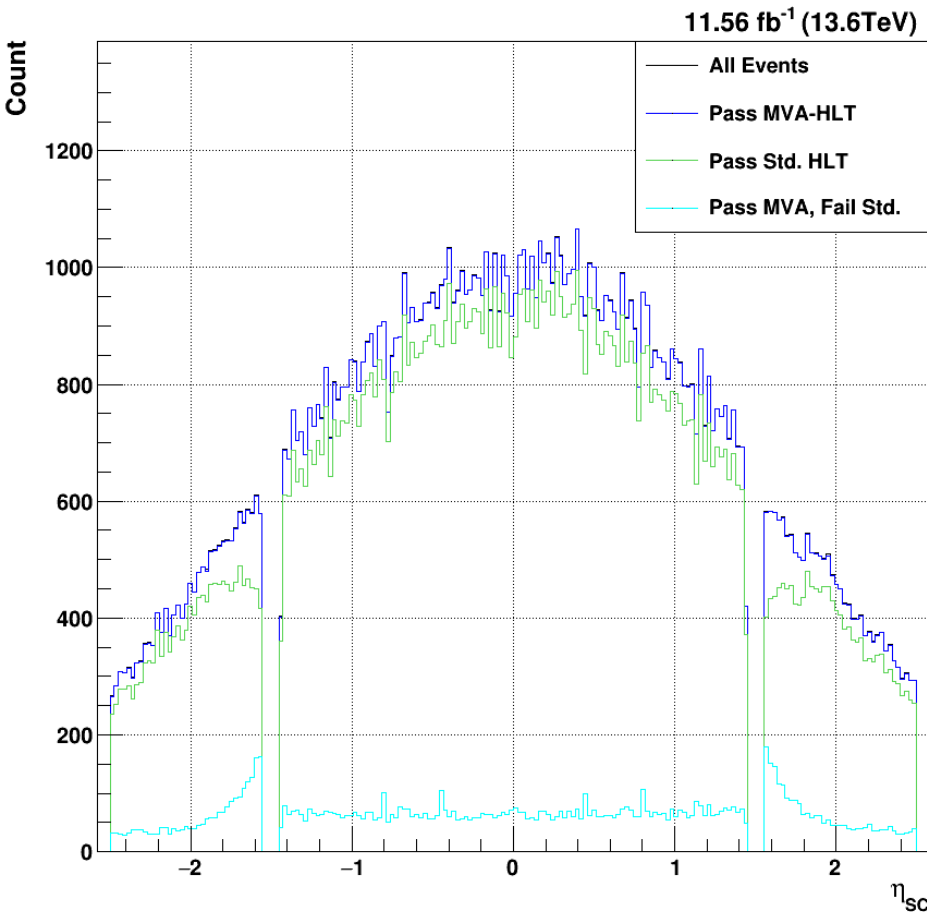
- The data used must pass any of the e/γ HLTs (different from ephemeral).
- Our goal is two-fold:
 - We must accept as much signal as possible.
 - We must not increase the data rate.

Single Higgs η_{SC}



Higgs MC Signal

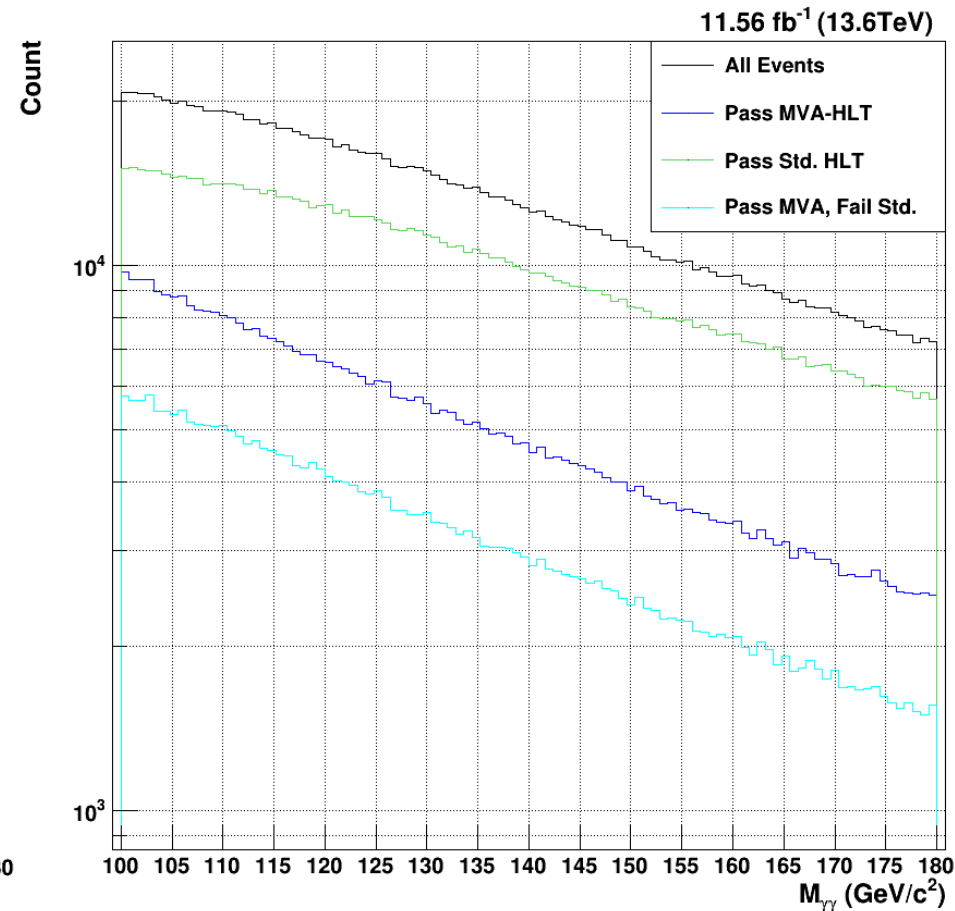
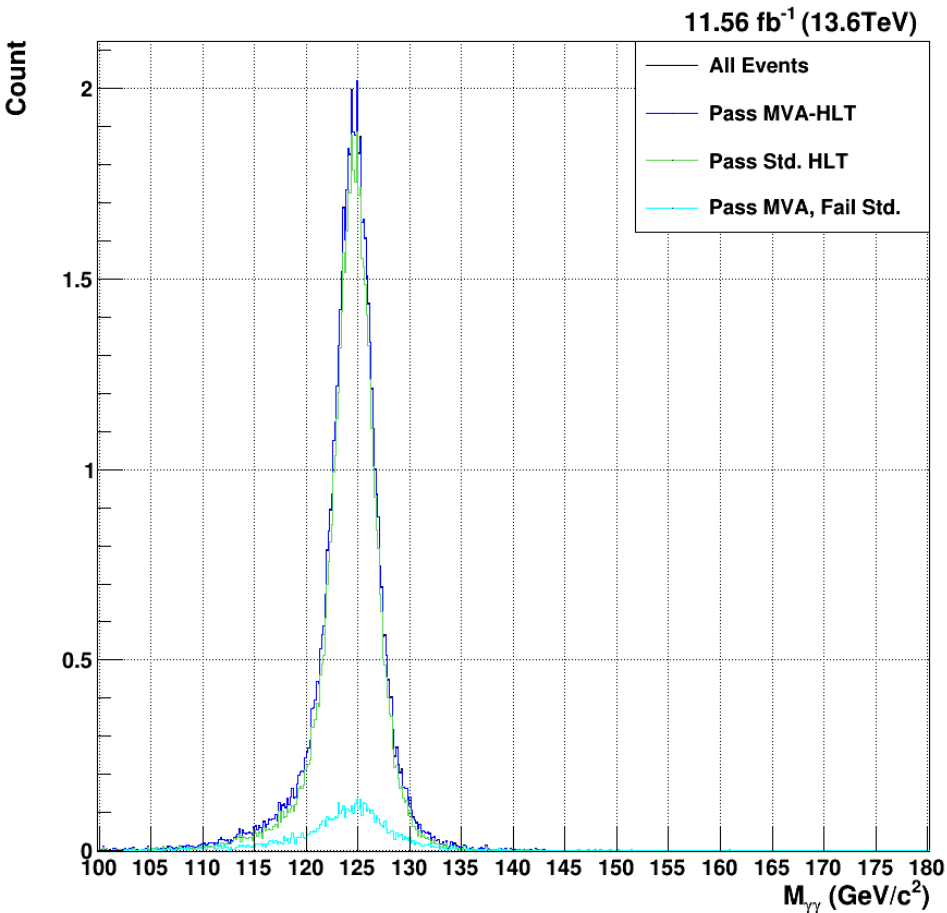
Data



- The **sHLT** shows a relative signal MC inefficiency at the start of the endcap.
 - Due to hard R_9 cuts
 - The **MVA-HLT** has consistent efficiency throughout the η_{SC} of the signal MC.
 - Improved kinematic coverage!
 - The **sHLT** accepts many background events with very high η_{SC} .
- The **MVA-HLT** improves the kinematic coverage in η_{SC} relative to the **sHLT**.

DiHiggs MC Signal

Data



- The signal efficiency of the **MVA-HLT** (98.5%) is higher than that of the **sHLT** (88.5%).
- The **MVA-HLT** generalizes well to the DiHiggs sample.

- The signal MC and data here have additional requirements on the second Higgs boson (these additional cuts have been established by the $HH \rightarrow b\bar{b}\gamma\gamma$ analysis group).
- Very similar performance is seen to that of the single Higgs.
 - This shows that the MVA-HLT generalizes well, even though it was trained on single Higgs signal MC.

Trigger-Level Efficiency Summary



Single Higgs

η_{SC} Region	sHLT Efficiency	MVA-HLT Efficiency	Efficiency Gain
B-B	93.39	98.65	+ 5.6
B-E	81.27	95.20	+ 17.1
E-E	86.24	96.70	+ 12.1
ALL	88.68	97.32	+ 9.7

DiHiggs

η_{SC} Region	sHLT Efficiency	MVA-HLT Efficiency	Efficiency Gain
B-B	91.01	98.90	+ 8.7
B-E	81.10	97.25	+ 19.9
E-E	89.18	98.26	+ 10.2
ALL	88.45	98.45	+ 11.3

- The total signal efficiency gain is appreciable for both the single Higgs and DiHiggs MC samples.
- In both cases, this accompanies a much lower trigger rate.
- So, the MVA-HLT is able to accept fewer data events overall, while still accepting more signal events. This will lead to an overall improvement to the signal purity of the data.
- To really leverage and improve upon these gains, however, changes must be made to the subsequent offline selection.

Development of a BDT-Based Offline Selection Optimized in Conjunction with the MVA-HLT

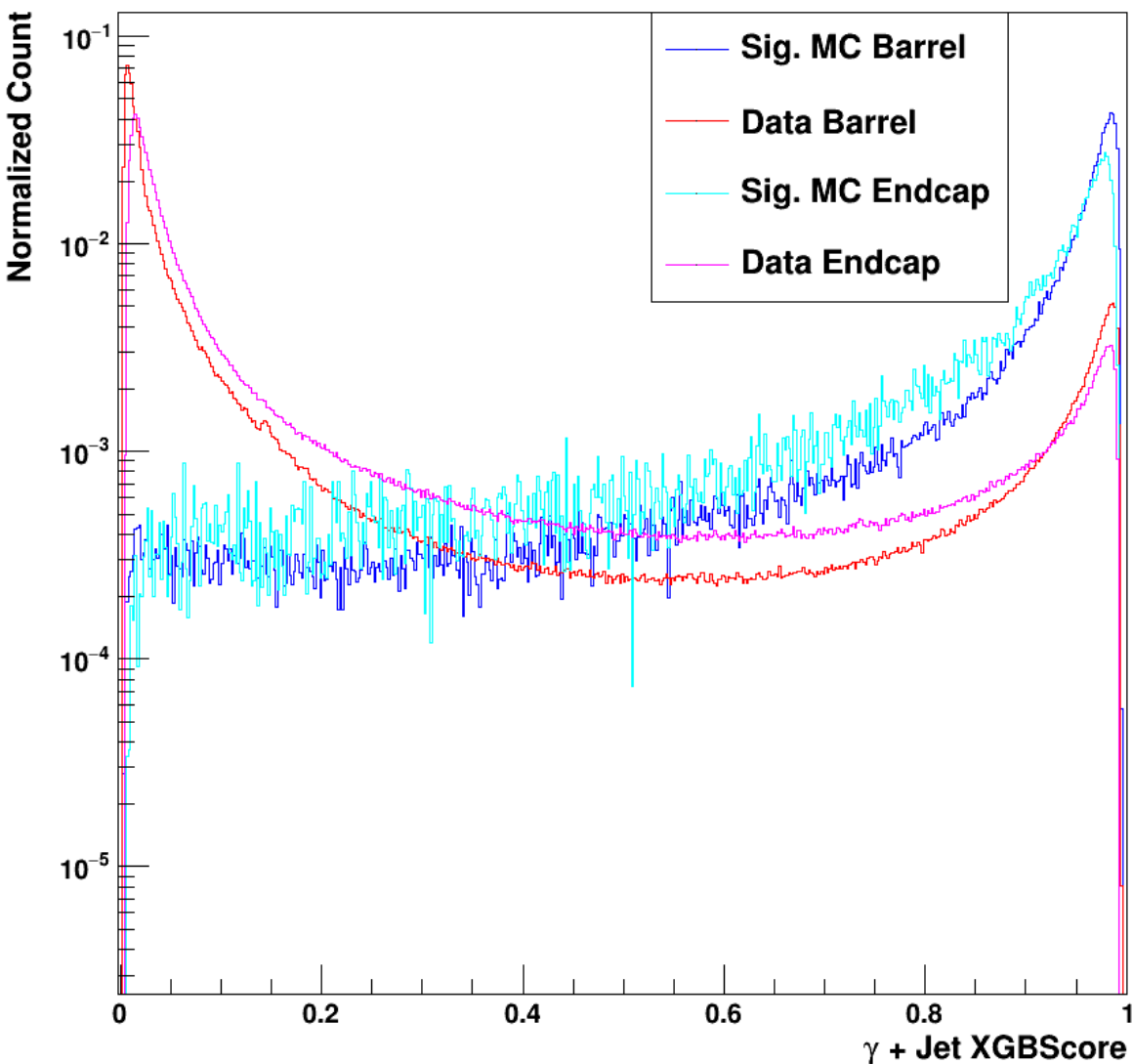


- The HLT is primarily designed to reduce the overall data processing rate while preserving as many important signal events as possible.
- Physics analysis and measurements occur after the HLT is applied to the data, when these samples are processed and final “offline selections” are applied to the events passing the trigger.
- The HLT-level cuts mirror the eventual offline analysis cuts.
 - Due to improved reconstruction and corrections, the offline analysis cuts are tighter than the HLT cuts.
- In order to leverage the gains made by the MVA-HLT, a new offline selection must be developed with a similarly high signal efficiency.
 - As with the Standard Analysis (SA) cuts and the sHLT, this new selection should be constructed on similar principles to the MVA-HLT.



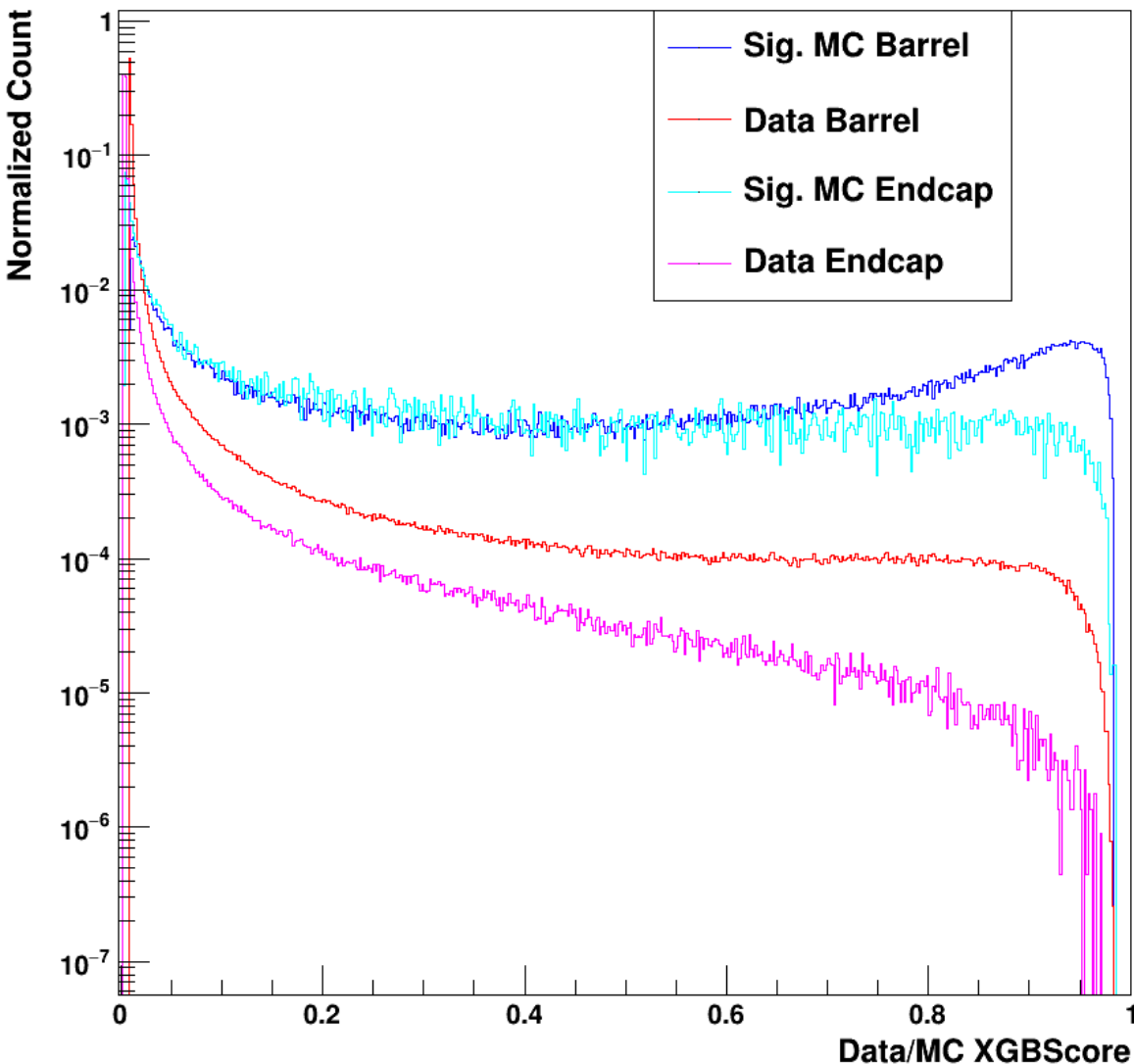
- The standard analysis (SA) cuts have historically employed loose cuts on a basic single photon BDT to reject poor photon candidates.
- Hadronic “Jets” are a major source of background and must be cut out at the analysis level.
- CMS has used real and fake photons from a γ +jet event to train photonID.
 - Train with γ (prompt) as signal, fake as background.
 - Fakes from jets or $\pi^0 \rightarrow \gamma\gamma$, for example.
- We have trained a new version of this γ +jet BDT for the BDT-based offline analysis.
 - We use the same set of variables used to make the preselection cuts, just as we did for the MVA-HLT.
 - As with the MVA-HLT, the hyperparameters of this BDT were experimentally optimized to maximize the performance of the model on the selected sample.

DiHiggs γ +jet BDT Scores



- The BDT scores of the Signal (Barrel and Endcap) and Data (Barrel and Endcap) are plotted together on a semi-log scale.
- Clearly, the signal MC has scores mostly near 1 (good).
- The data also have many scores near 1 (bad).
- Since the data has many real photons in it, the γ +jet BDT cannot eliminate all of the background in the data.

DiHiggs Data/MC BDT Scores

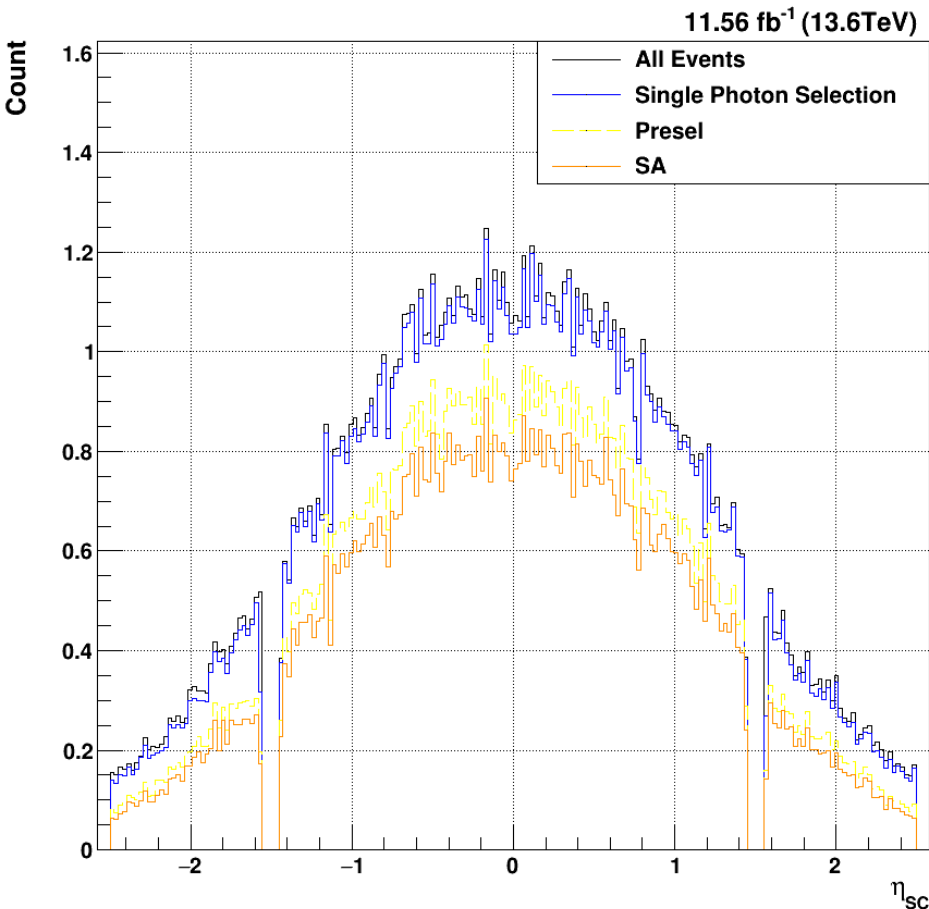


- We created a new BDT trained on signal MC and Data (Data/MC BDT)
 - γ +jet BDT alone cannot remove photon backgrounds
 - More similar to the method of the MVA-HLT
- The Data/MC shows the opposite issue to the γ +jet BDT; while the data is very well-localized near 0, there is still a lot of signal given a score close to 0.
- We have trained different version of this BDT for the Single and DiHiggs samples.

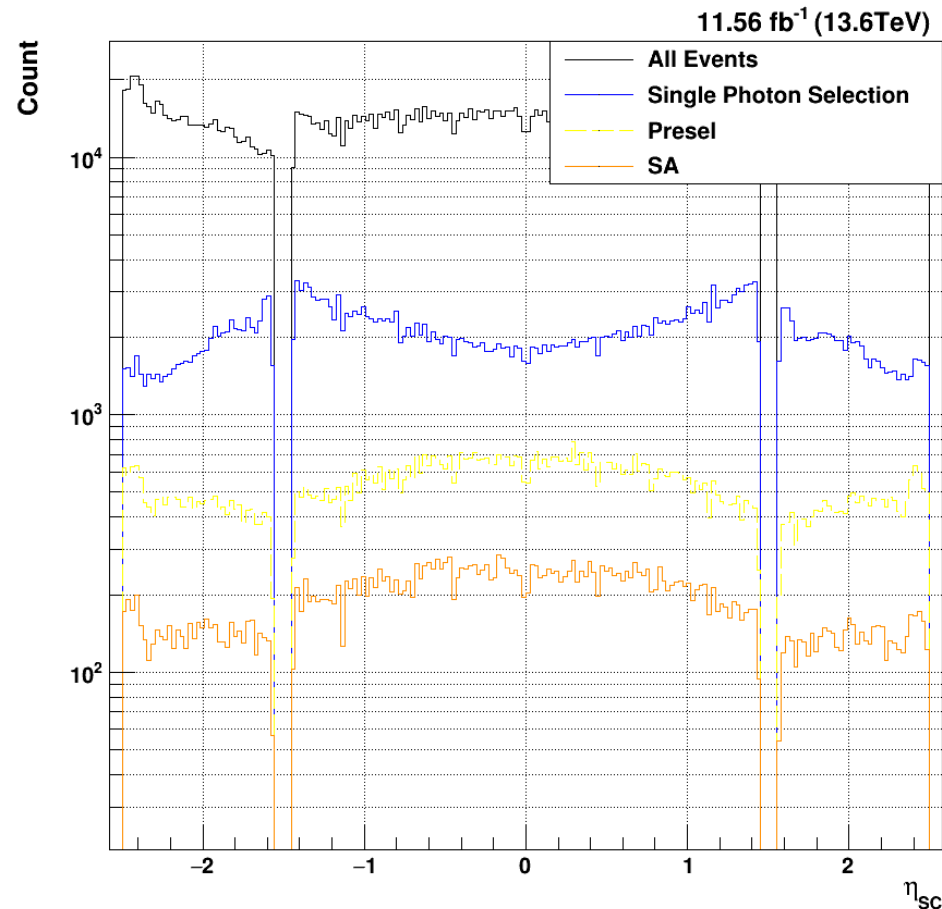
DiHiggs Single Photon BDT Cuts



DiHiggs MC Signal



Data



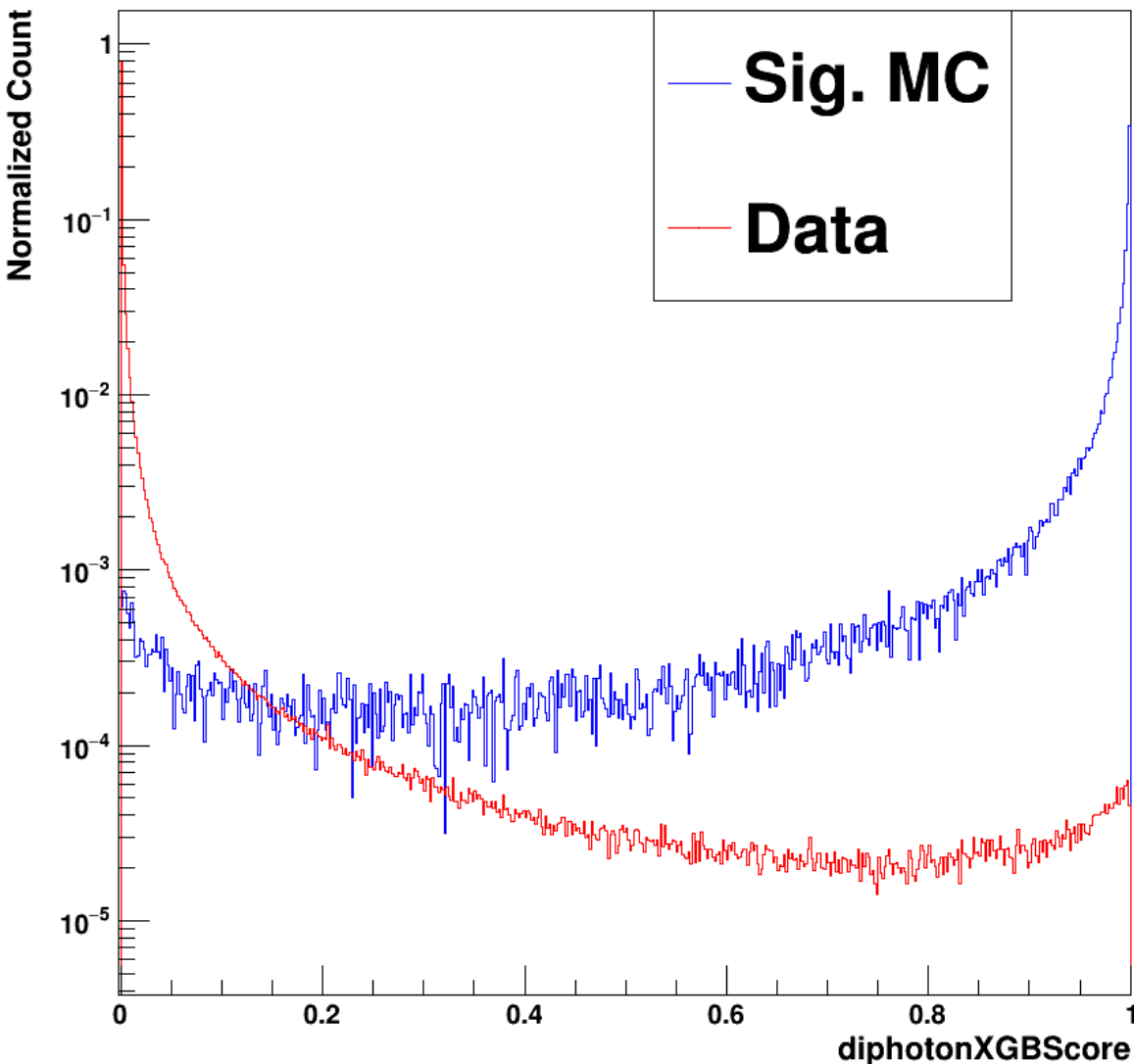
- Compared to the single photon **Preselection** and full **Standard Analysis (SA)** cuts, these **loose single photon cuts** have much higher signal efficiency but also accept more background.

- Very loose cuts** can be applied to reduce the data rate while still keeping the signal efficiency very high (>98%).
- Cuts on single photons are not adequate to fully reduce the data rate, as there are many real diphoton events in the data.



- So far, both offline BDTs developed have been for selecting individual photon candidates.
- The SA cuts, however, make cuts on the $p_T/M_{\gamma\gamma}$, which is a diphoton quantity.
- This suggests the creation of a BDT that uses information from both photons together to select events instead of individual photons – a diphoton BDT.
- Run 2 analyses at CMS made use of a diphoton BDT for event categorization, so this marks a good starting point for the fully BDT-based approach.
 - Diphoton geometric quantities, $p_T/M_{\gamma\gamma}$ of each photon, and the single photon BDT scores of each photon are used as input variables.

DiHiggs Diphoton BDT Scores



- Much better separation between signal and data can be seen for the diphoton BDT than either single photon BDT.
- A very loose cut on this score can clearly reduce the data rate considerably while still minimizing the loss of signal efficiency.
- This new strategy leverages both single photon and diphoton information, mirroring the SA selection.
- A very high signal efficiency is expected from cuts on this BDT score.

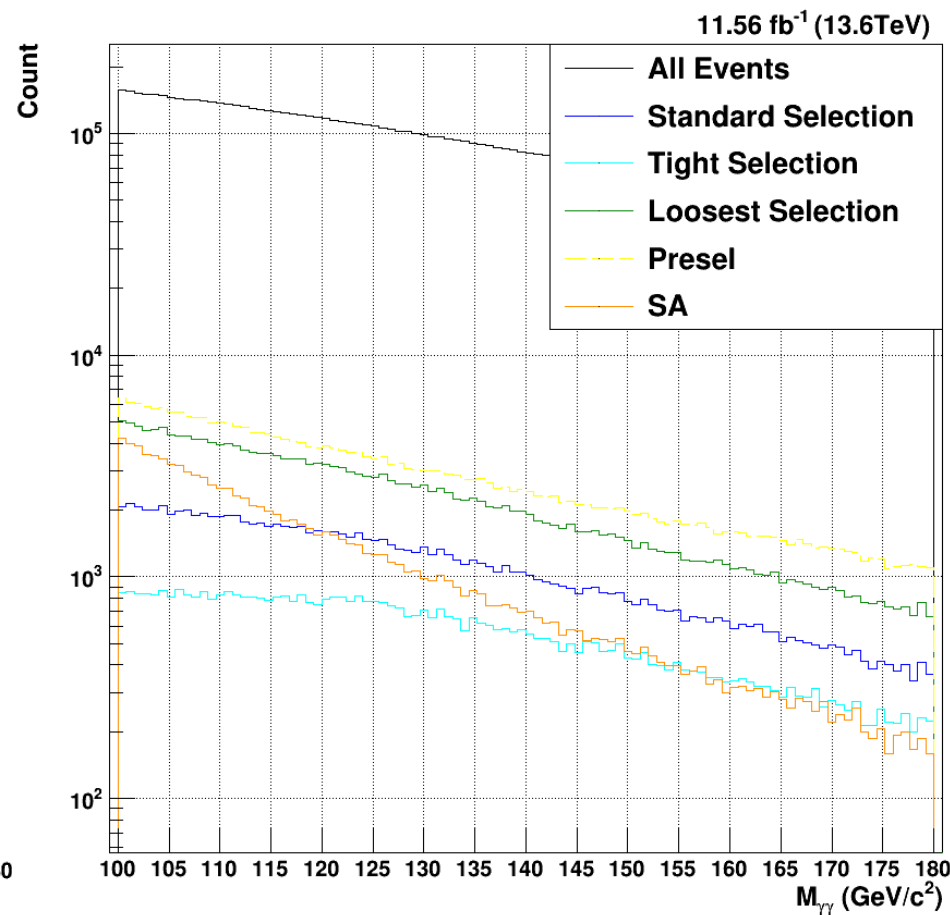
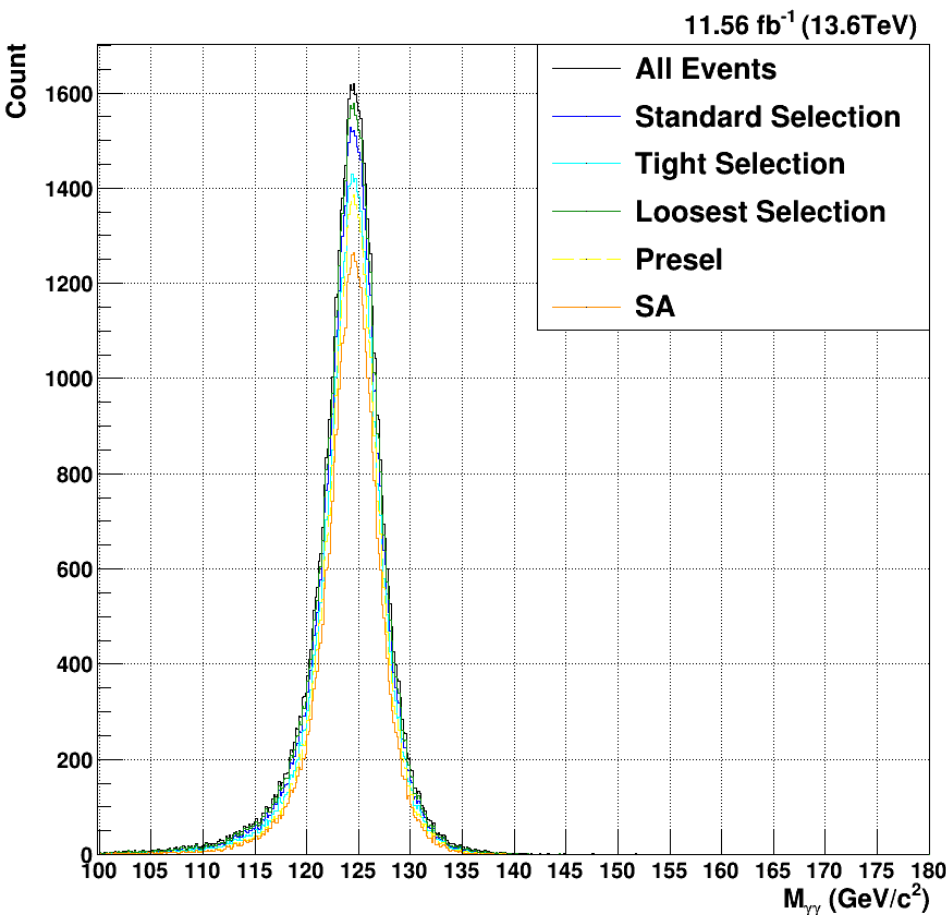
Full Offline Selection Results for $H \rightarrow \gamma\gamma$ and $HH \rightarrow b\bar{b}\gamma\gamma$

Single Higgs $M_{\gamma\gamma}$



Higgs MC Signal

Data

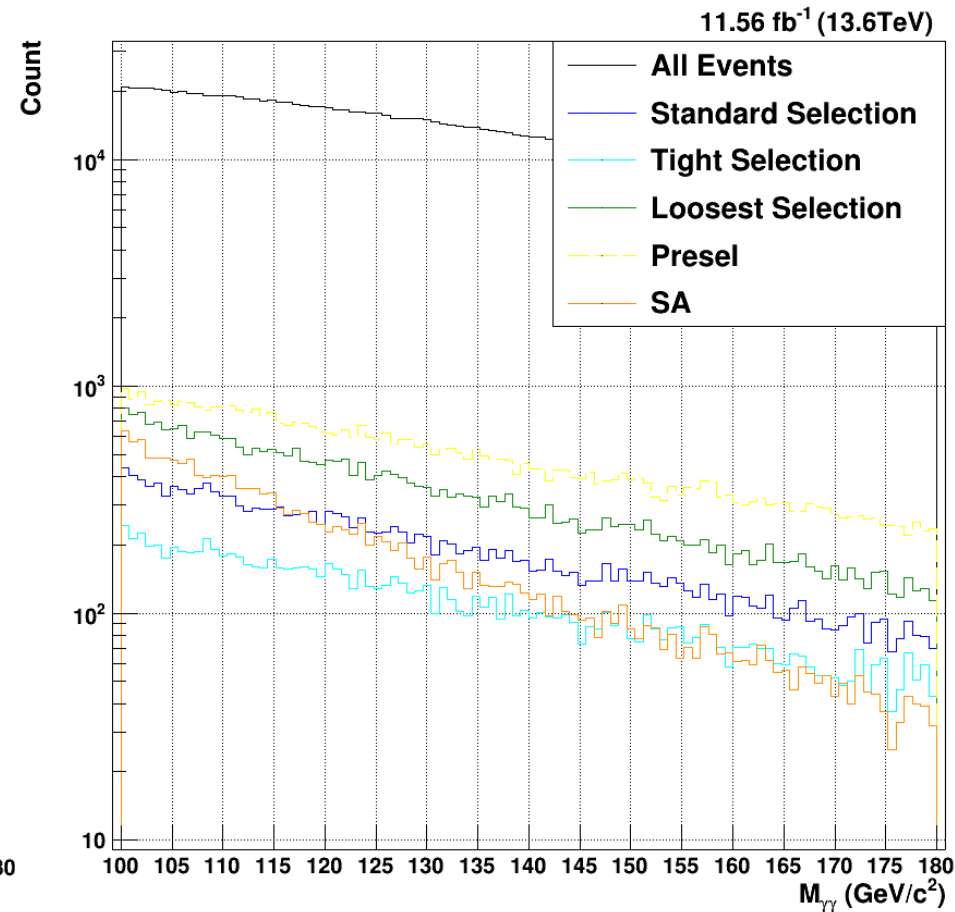
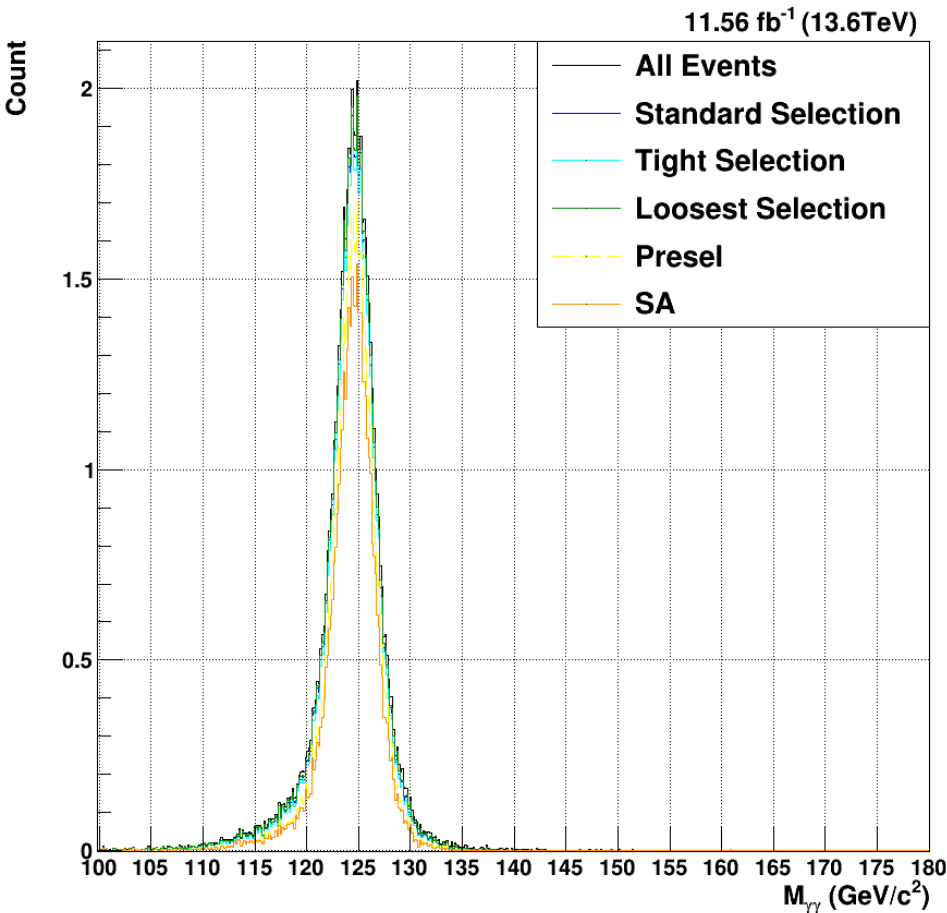


- With the same amount of background, the BDT-Based selection achieves a 29.4% gain in signal efficiency
- The background mass distribution is affected by the SA cuts.

- To compare to the **Standard Analysis** and **Preselection** cuts, diphoton BDT cuts must be chosen and optimized.
- The **Loosest cuts** accept 2x as much background as the SA.
- The **Standard Selection** cuts accept the same amount of background as the SA
- The **Tightest cuts** accept ½ as much background as the SA.
- These cuts are chosen as an example of an analysis strategy that preserves high signal efficiency, but each analysis has freedom to choose their own strategy.

DiHiggs MC Signal

Data



- With the same amount of background, the BDT-Based selection achieves a 39.7% gain in signal efficiency.
- The BDT-based selection works well for the DiHiggs.

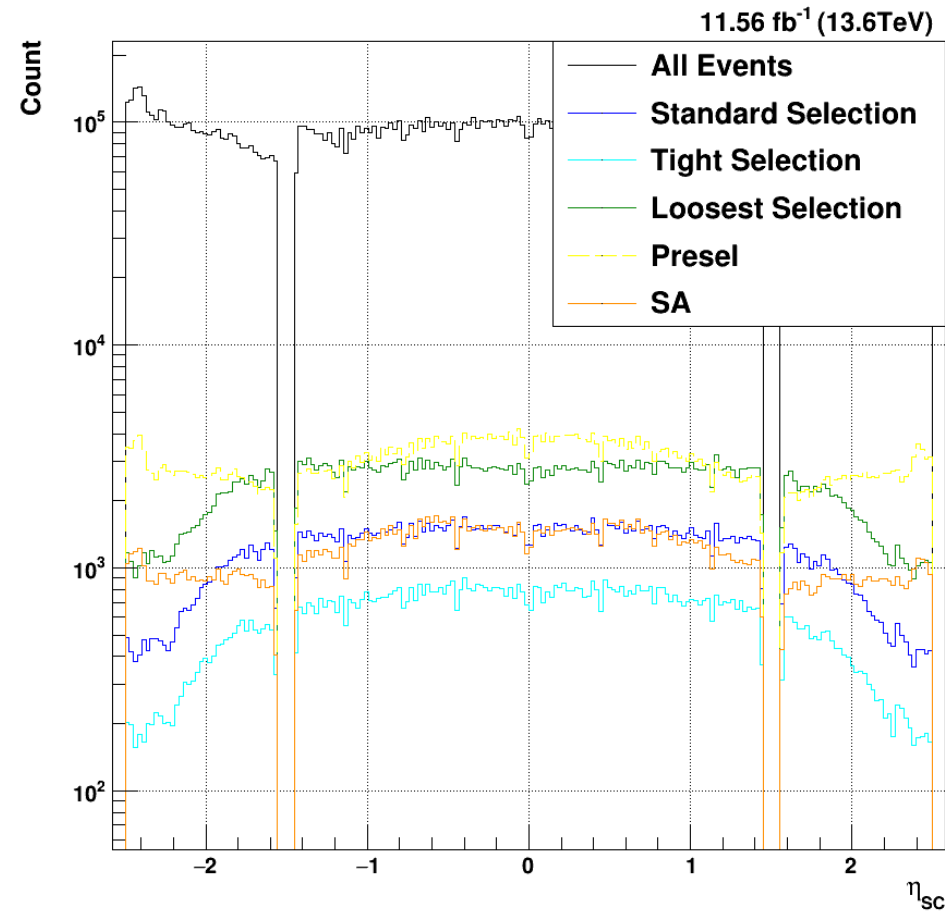
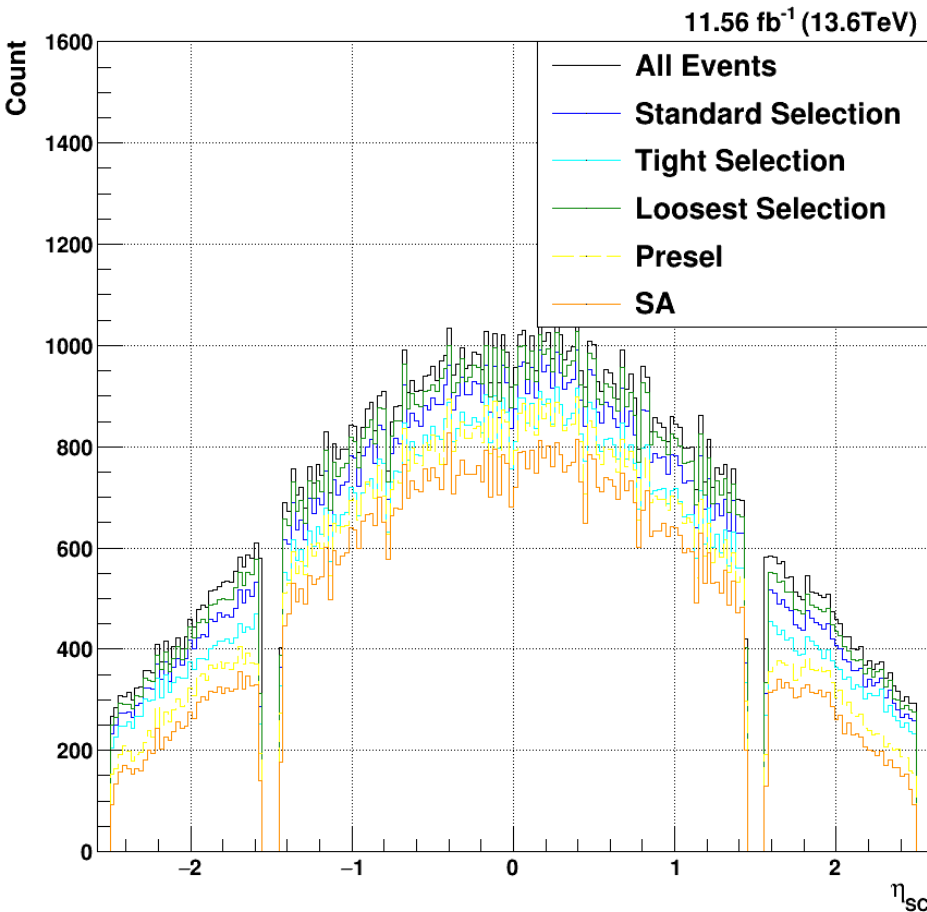
- The DiHiggs mass distribution is relatively similar for both the signal MC and data.
- The overall signal efficiency is improved for the DiHiggs (94.7%) compared to the Single Higgs (91.3%).
- Eventual analyses will fit the background and signal mass distributions, so offline selections must support the fitting procedure by not shaping the signal or background mass shape.

Single Higgs η_{sc}



Higgs MC Signal

Data



- In addition to the overall improvement to signal efficiency, the kinematic coverage is improved by the BDT-based offline selection.

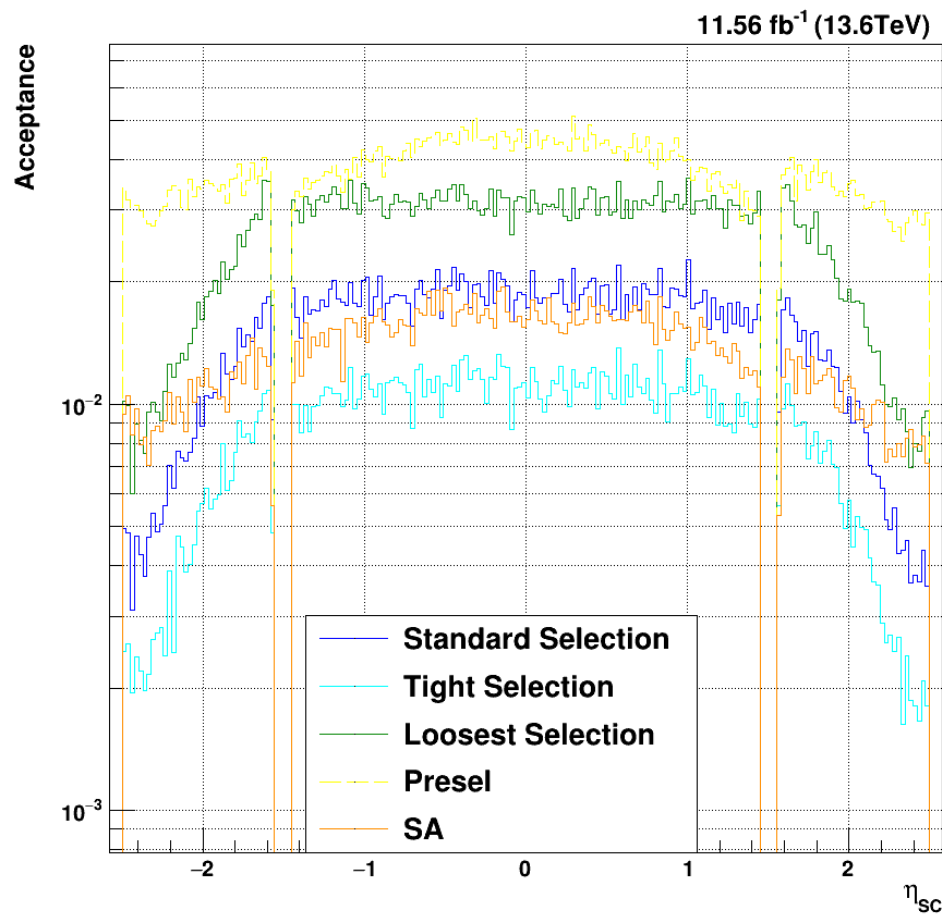
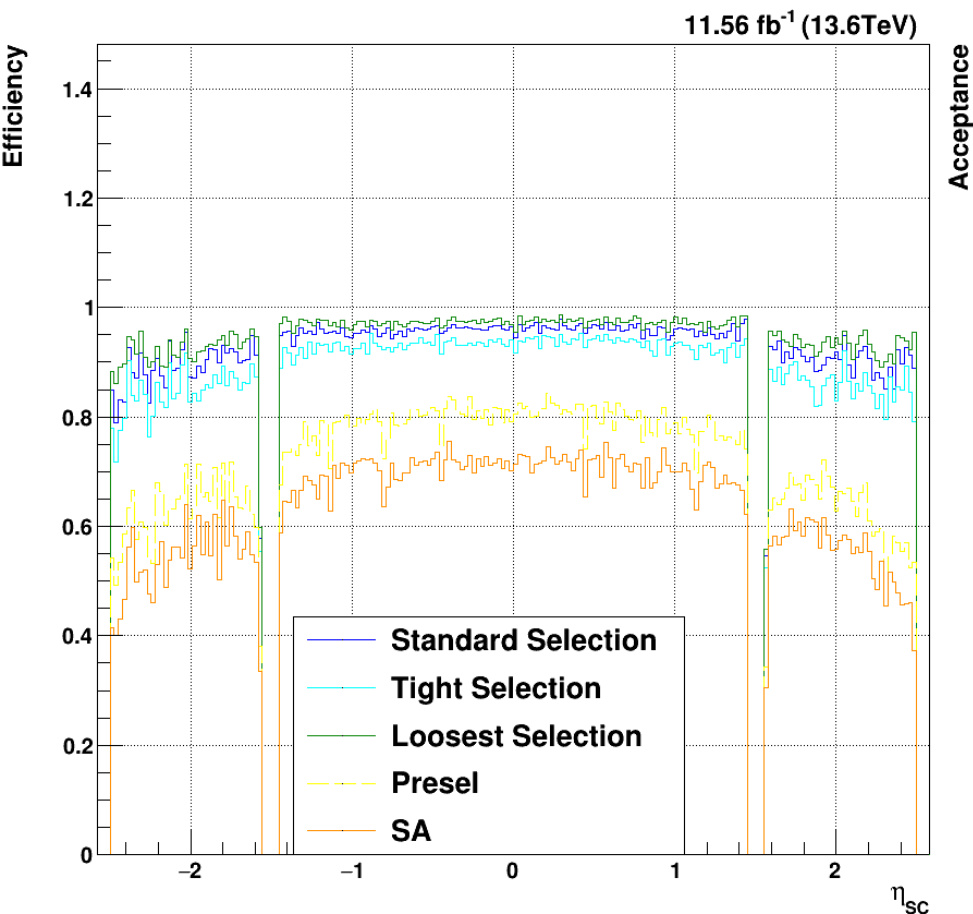
- As with the sHLT, the presel and SA cuts reduce the efficiency of the signal at the lower-eta region of the endcap.
- The BDT-based selections all tend to accept less data in the higher-eta region of the endcap.

DiHiggs η_{SC} Efficiency



DiHiggs MC Signal

Data



- The BDT-based selection keeps relatively flat efficiency throughout the selected η_{SC} range.

- The signal inefficiencies of the SA cuts are clearer here.
- Again, the BDT-based selections reduce the acceptance of high- η data, a region that has very little signal.
- Since the preselection and SA cuts are designed to “cover” the sHLT, we must remove these cuts to leverage the improved signal efficiency of the MVA-HLT.



Single Higgs

η_{SC} Region	SA Efficiency	BDT Efficiency	Efficiency Gain
B-B	81.95	94.56	+15.4
B-E	52.12	85.43	+63.9
E-E	63.63	91.63	+44.0
ALL	70.53	91.27	+29.4

DiHiggs

η_{SC} Region	SA Efficiency	BDT Efficiency	Efficiency Gain
B-B	73.43	97.23	+32.4
B-E	53.83	88.38	+64.2
E-E	57.02	89.88	+57.6
ALL	67.78	94.68	+39.7

- Huge gains are achieved at the selection level for both the single and DiHiggs signal MC.
- The gains are largest for events where one or both photons are in the barrel, showing the increase in kinematic coverage.
- A selection efficiency above 90% with greatly benefit both single and DiHiggs analyses.
- This gain is achieved by changing both the HLT and the offline selection – must go hand-in-hand for maximum effect!



- A new BDT-based approach has been developed to improve the HLT and offline selection for $H \rightarrow \gamma\gamma$ events at the CMS experiment.
- The MVA-HLT is able to improve the signal efficiency of both $H \rightarrow \gamma\gamma$ and $HH \rightarrow b\bar{b}\gamma\gamma$ MC samples while still cutting the rate of accepted data almost in half.
- The complementary offline selection is able to improve the signal efficiency by 29% for the single Higgs and 39% for the DiHiggs compared to the standard analysis cuts.
- Looking forward, this new HLT and offline selection may prove a vital boon to future analyses, especially those targeting rare decays that include a $H \rightarrow \gamma\gamma$.



Thank You!



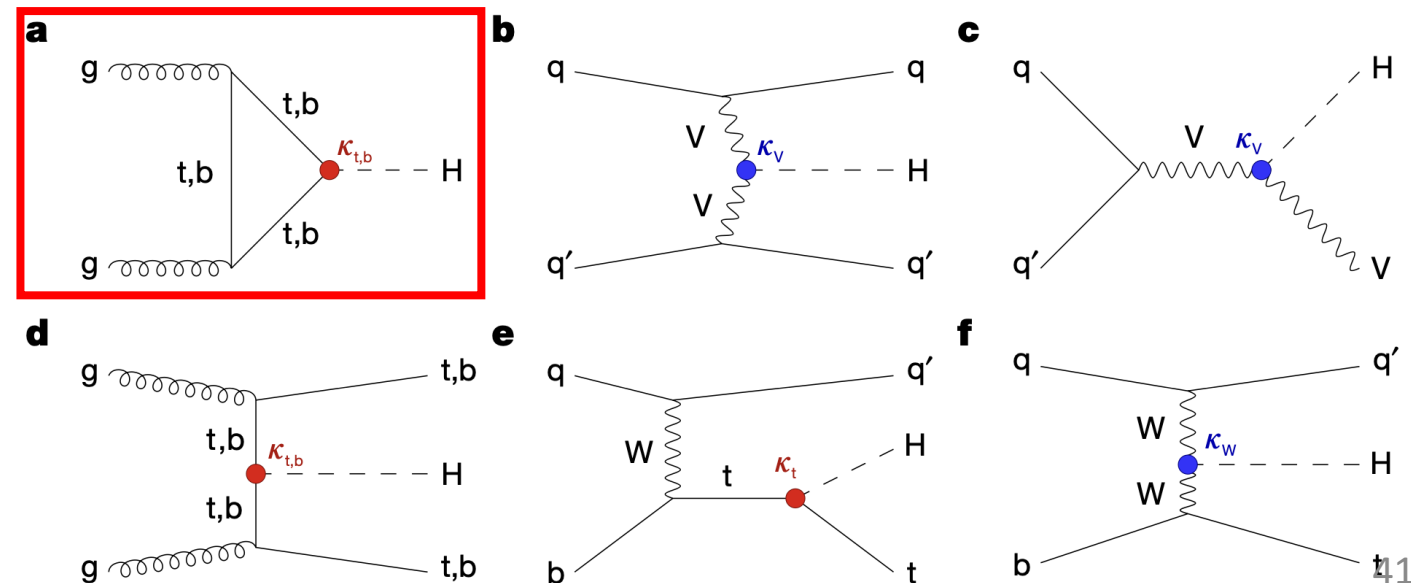
Backup

Single Higgs Boson Production Modes



- From the underlying field theories of interaction, Feynman diagrams can be constructed to describe the possible ways in which a Higgs Boson can be produced.
- "Cross Sections" measure how likely different Higgs production modes are. The total Higgs XS is $\sigma_H = 54 \pm 2.6$ pb:
 - a:** Gluon Fusion ≈ 48.3 pb
 - b:** Vector Boson Fusion ≈ 3.8 pb
 - c:** $VH \approx 2.2$ pb
 - d:** $tt(bb)H \approx 1.0$ pb
 - e&f:** $tH \approx 0.1$ pb

Higgs boson production modes

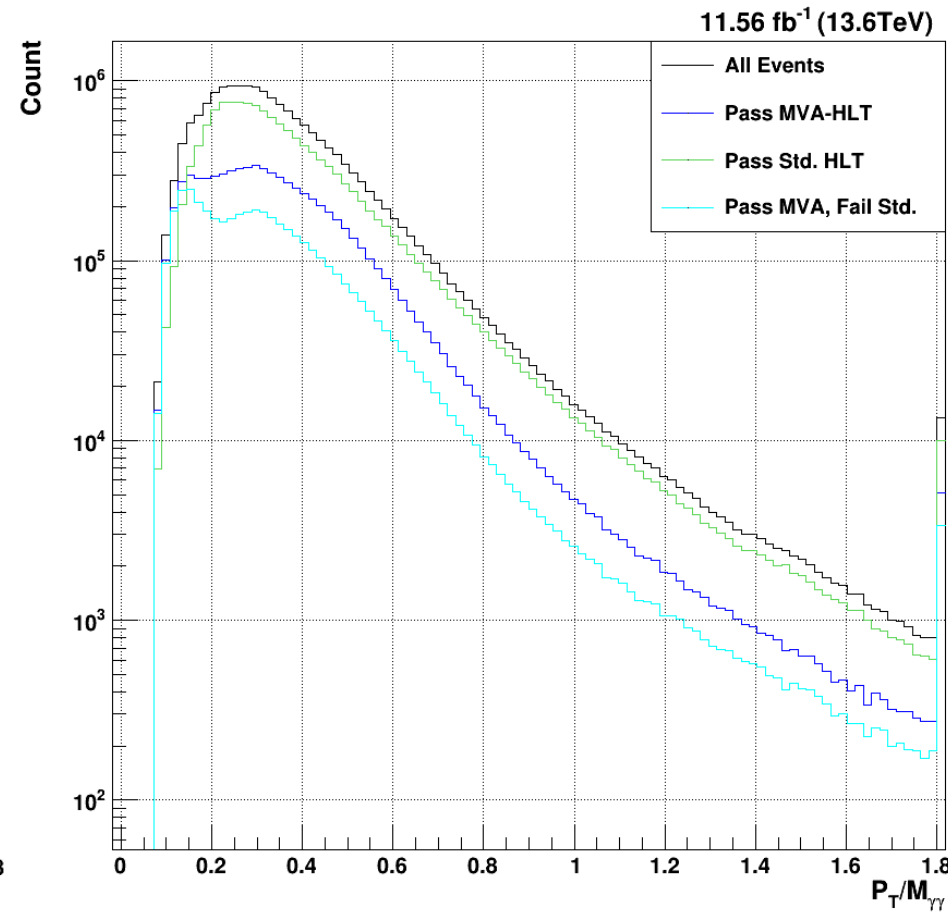
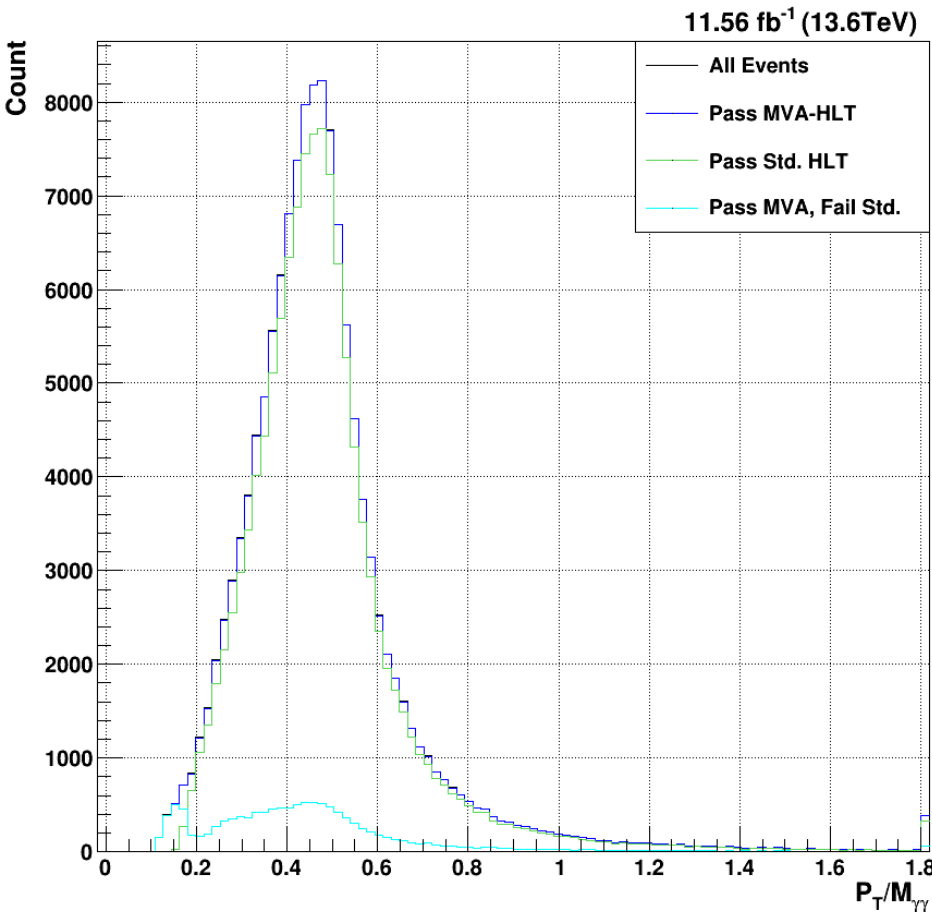


Single Higgs $p_T/M_{\gamma\gamma}$



Higgs MC Signal

Data



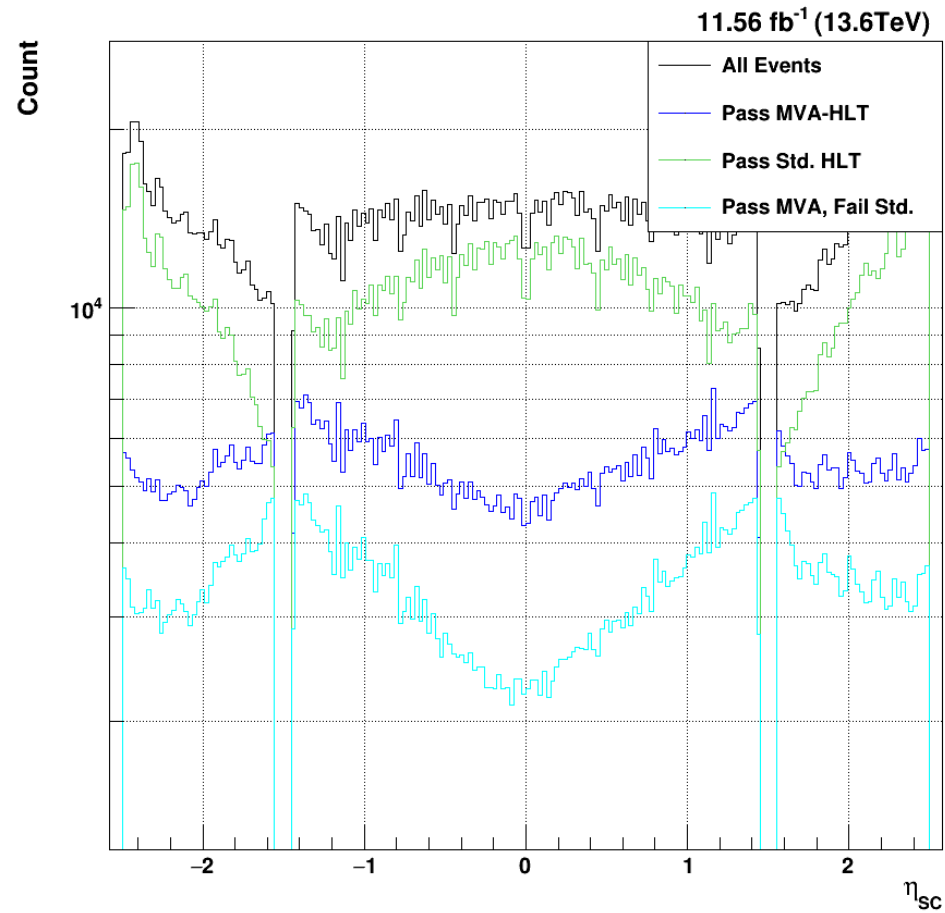
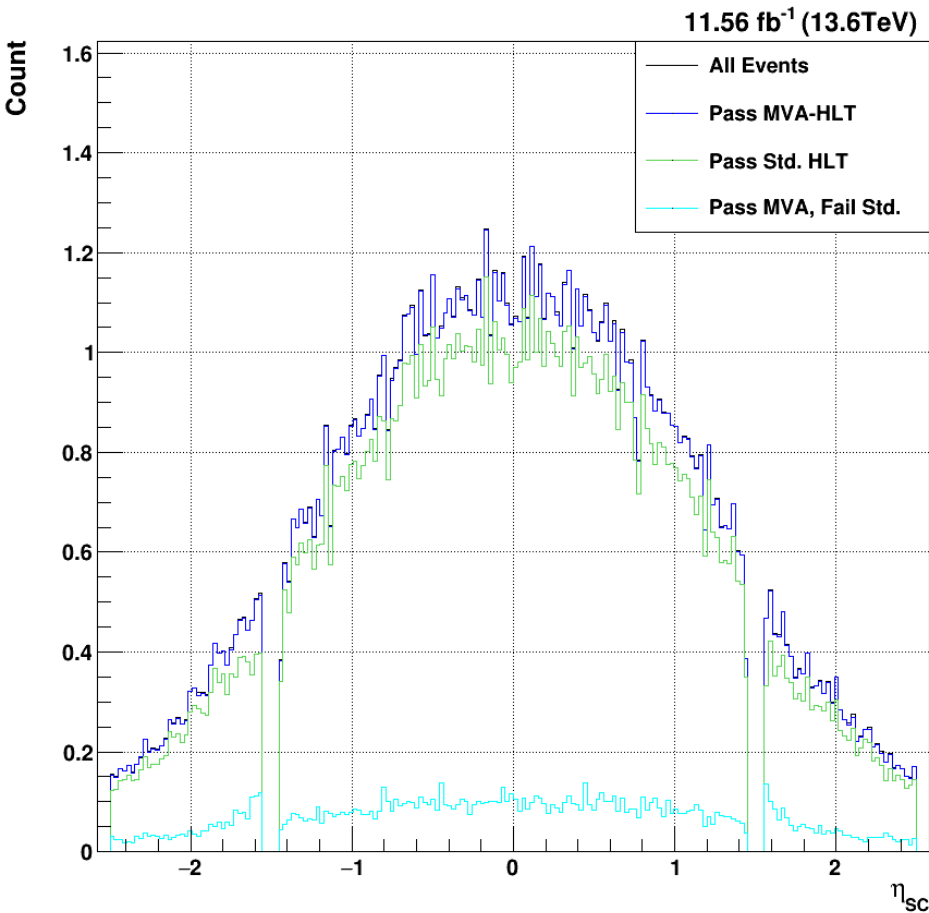
- The hard p_T cuts of the **sHLT** cut out signal events.
 - Without tight p_T cuts, the **MVA-HLT** gains back many of these events.
- The **MVA-HLT** improves the kinematic coverage in $p_T/M_{\gamma\gamma}$ relative to the **sHLT**.

DiHiggs η_{sc}



DiHiggs MC Signal

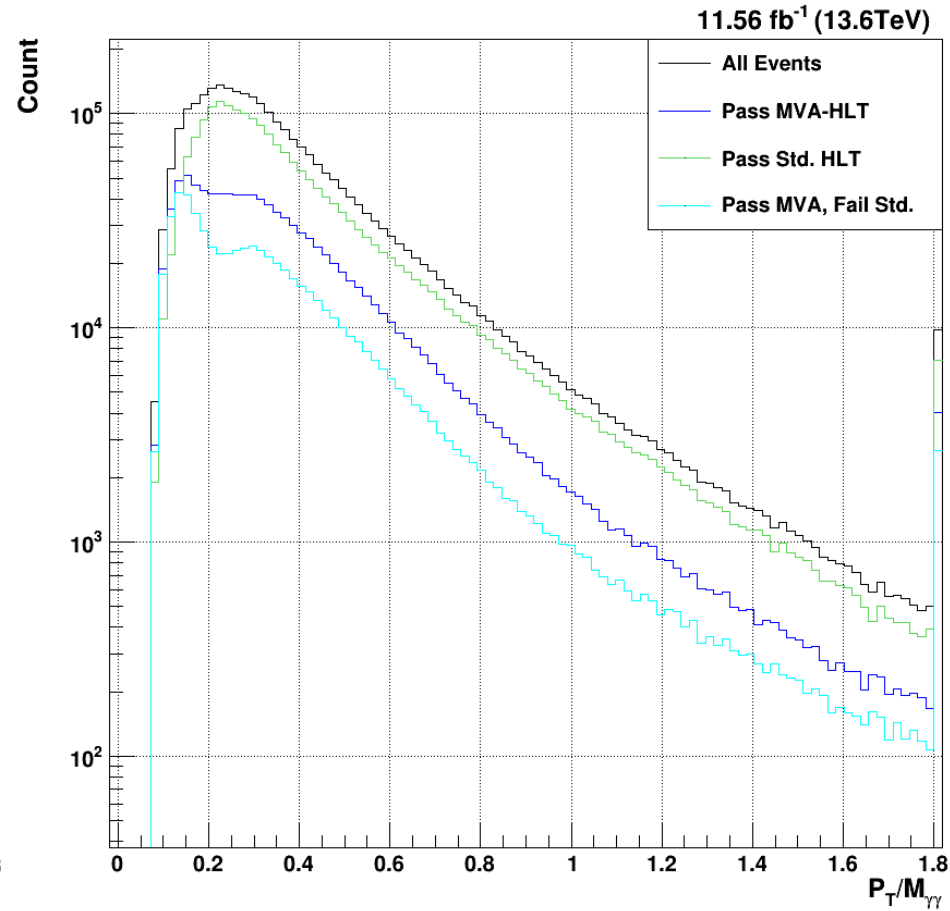
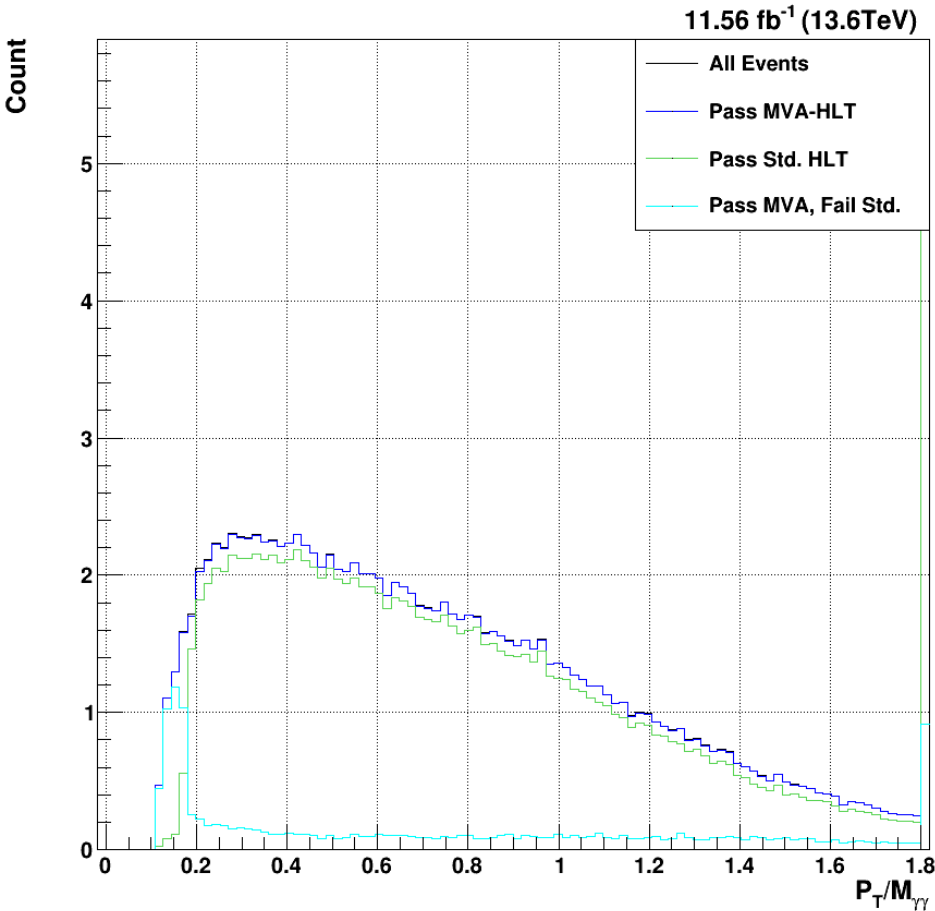
Data



- Similar improvements to the kinematic coverage can be seen for the DiHiggs sample.

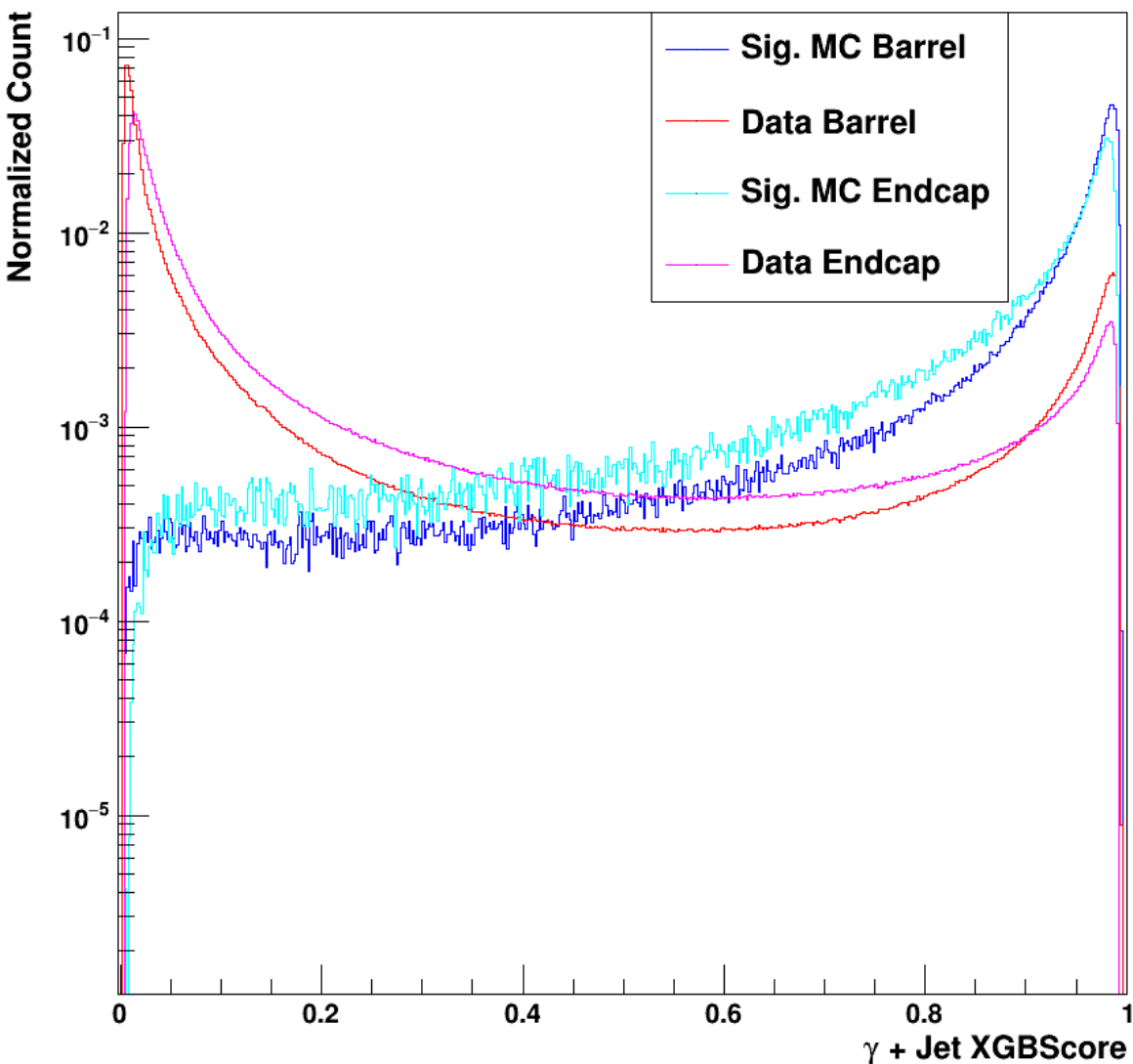
DiHiggs MC Signal

Data



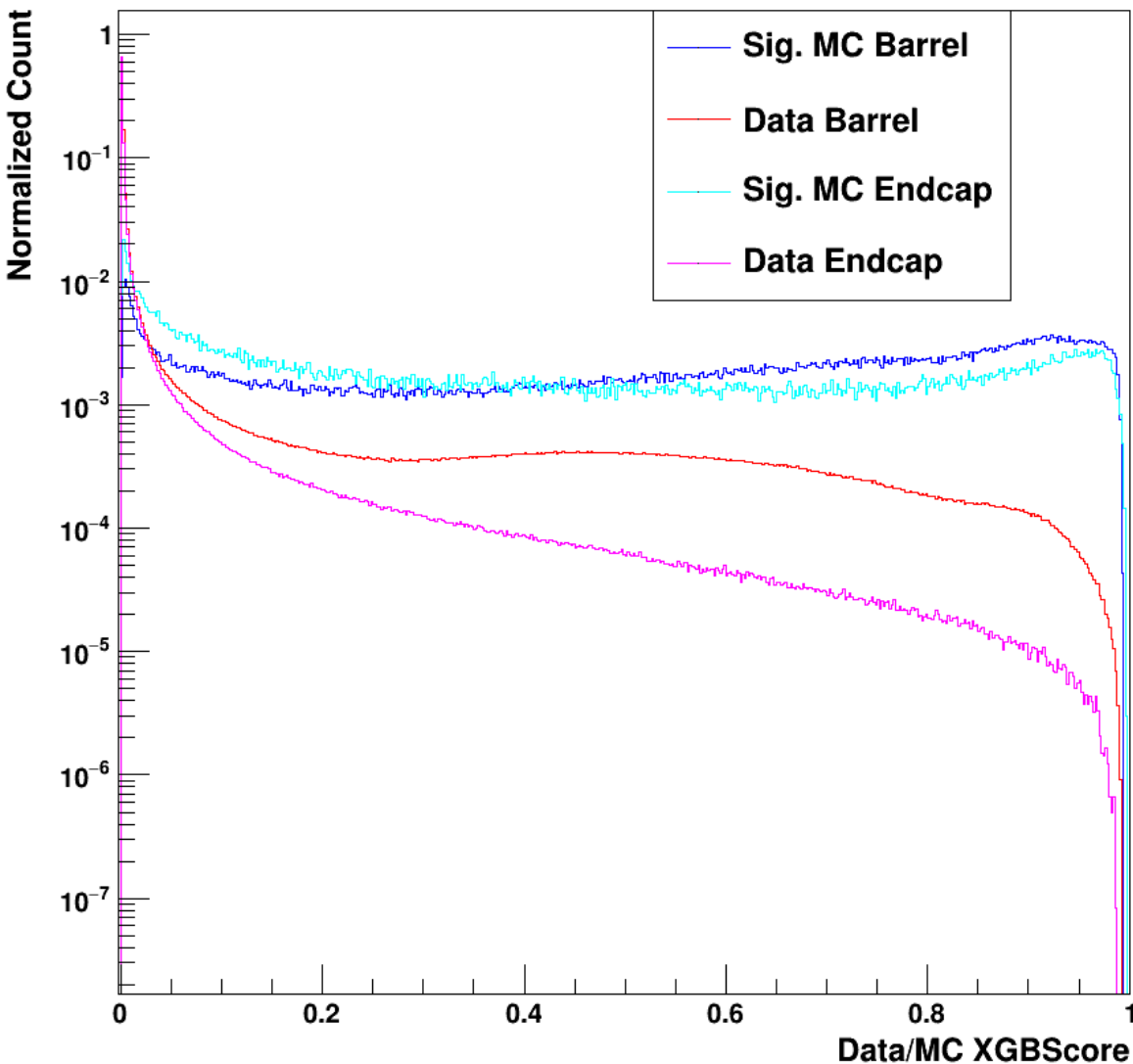
- The hard p_T cuts of the **HLT** have a stronger impact on the DiHiggs signal MC.
- The **MVA-HLT** is able to **gain more signal events** at lower $p_T/M_{\gamma\gamma}$.

Single H γ +jet BDT Scores



- The BDT scores of the Signal (Barrel and Endcap) and Data (Barrel and Endcap) are plotted together on a log plot.
- Clearly, the signal MC has scores mostly near 1 (good).
- The data, however, also has many scores near 1 (bad).
- Since the data has many real photons in it, the γ +jet BDT cannot adequately reduce the acceptance of the data.

Single H Data/MC BDT Scores



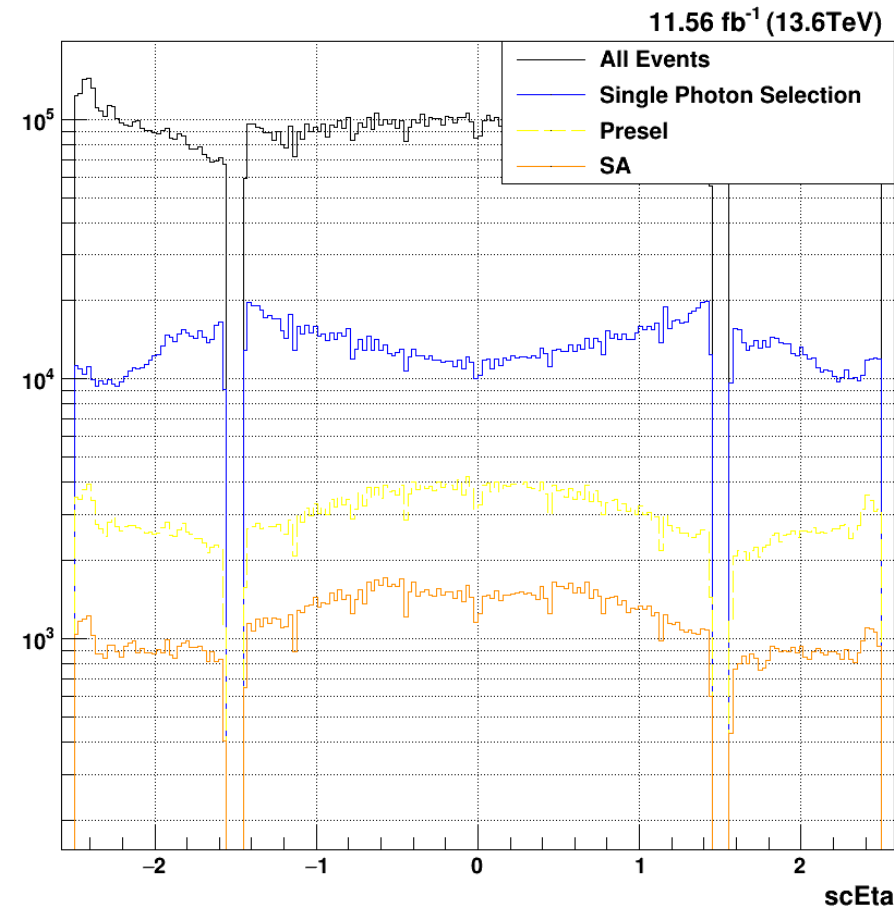
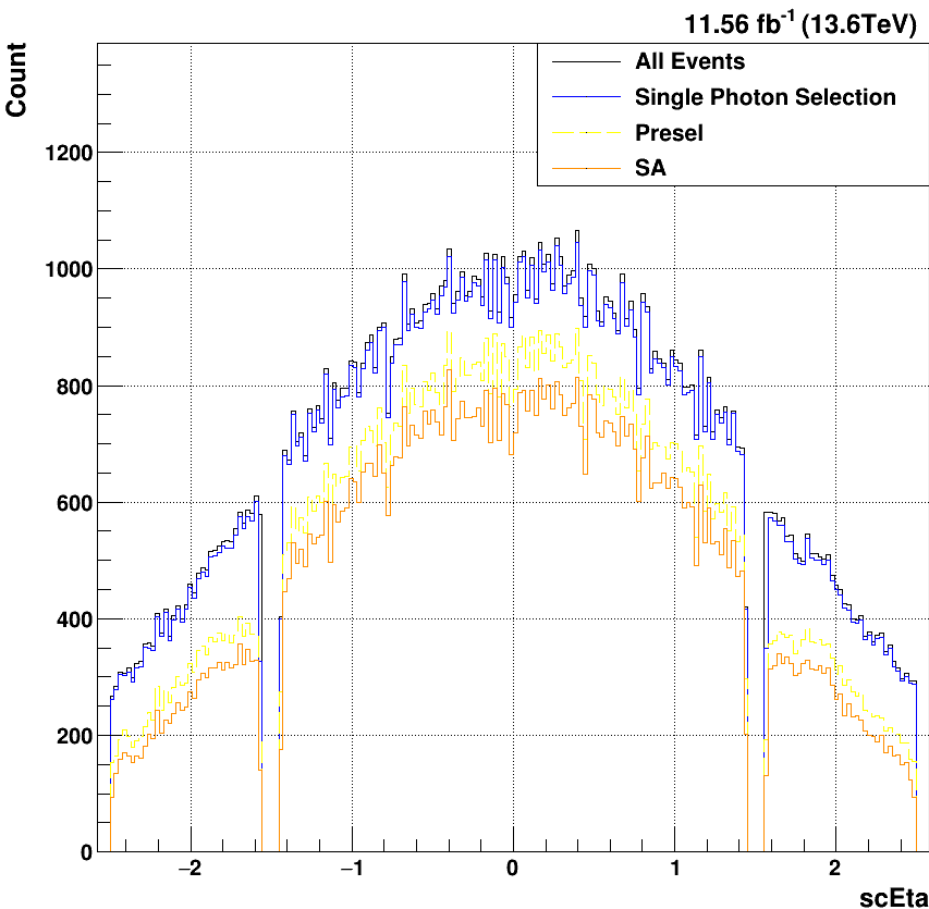
- The Data/MC shows the opposite issue to the γ +jet BDT; while the data is very well-localized near 0, there is still a lot of signal given a score close to 0.
- Cutting too tight on this score will significantly reduce the signal efficiency.
- Since the Data/MC BDT is trained using a specific signal MC sample, a different version is used for the Single Higgs.

Single Higgs Loose Single Photon BDT Cuts



Higgs MC Signal

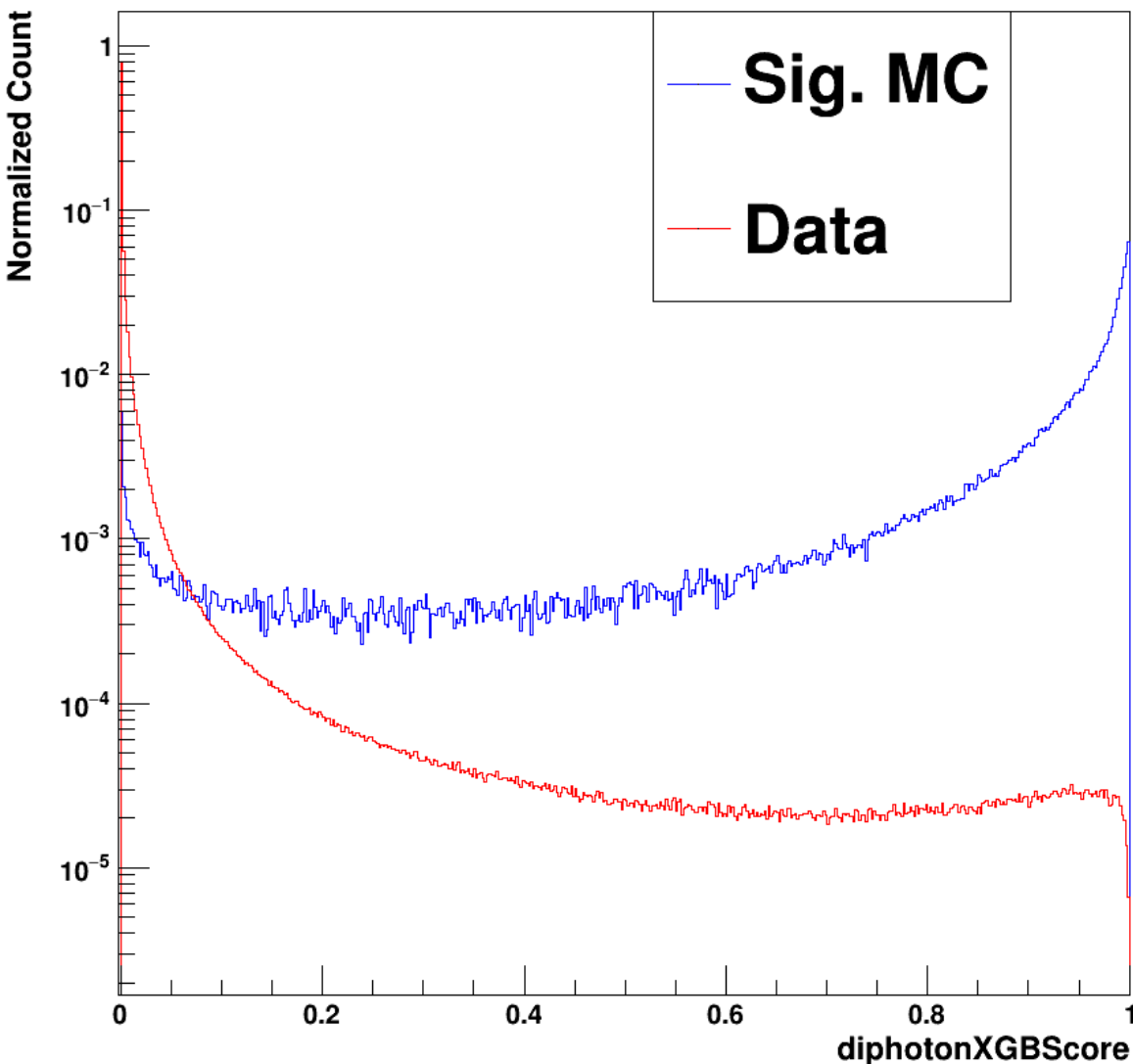
Data



- Instead of tight cuts on the single photon BDT scores, **very loose cuts** can be applied to significantly reduce the data rate while still keeping the signal efficiency very high (>98%).

- Compared to the single photon **Preselection** and full **Standard Analysis (SA) cuts**, these **loose single photon cuts** have much higher signal efficiency but also accept more background.

Single H Diphoton BDT Scores

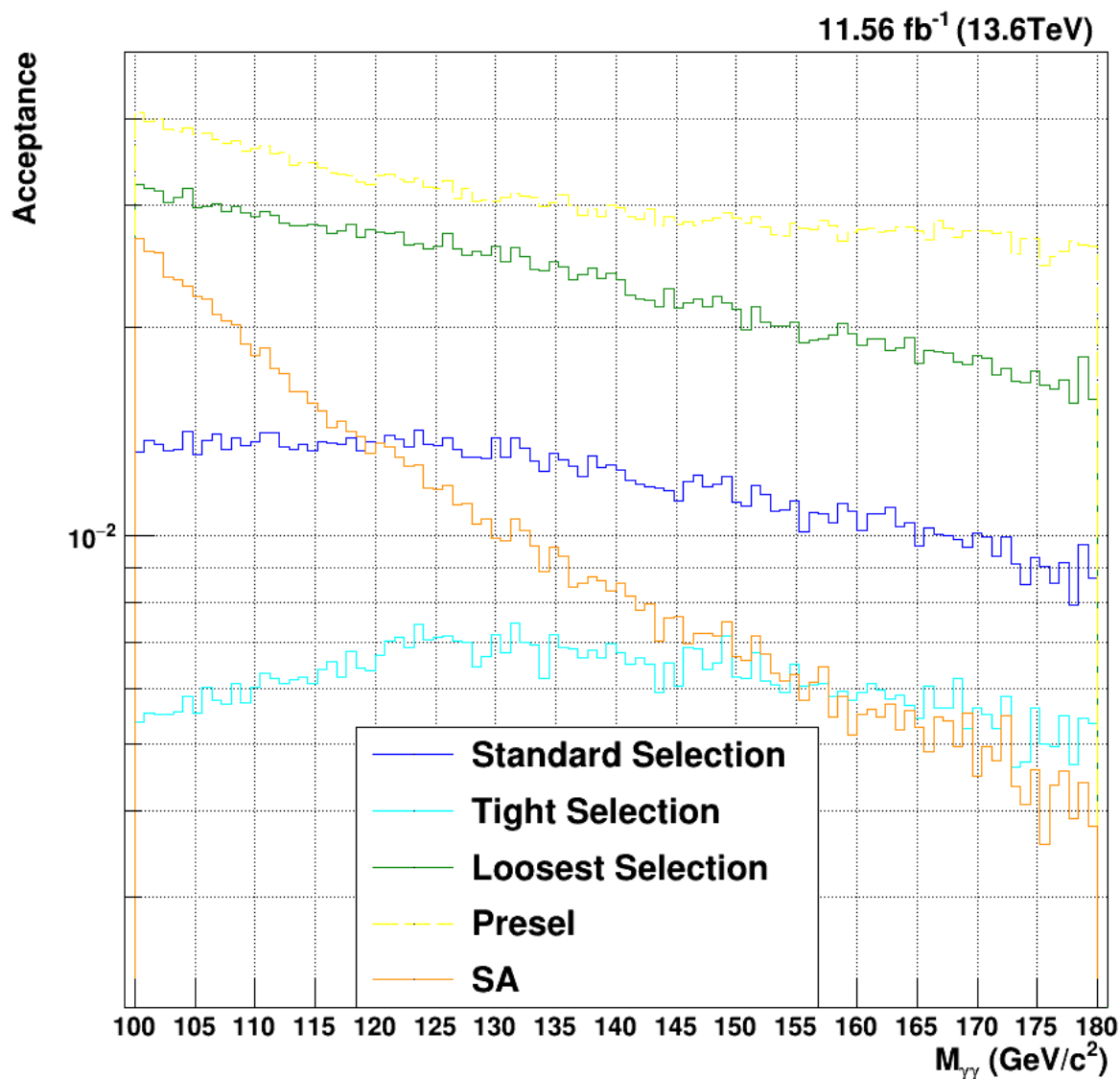


- Much better separation between signal and data can be seen for the diphoton BDT than either single photon BDT.
- A very loose cut on this score can clearly reduce the data rate considerably while still minimizing the loss of signal efficiency.
- This new strategy leverages both single photon and diphoton information, mirroring the SA selection.

Single Higgs $M_{\gamma\gamma}$

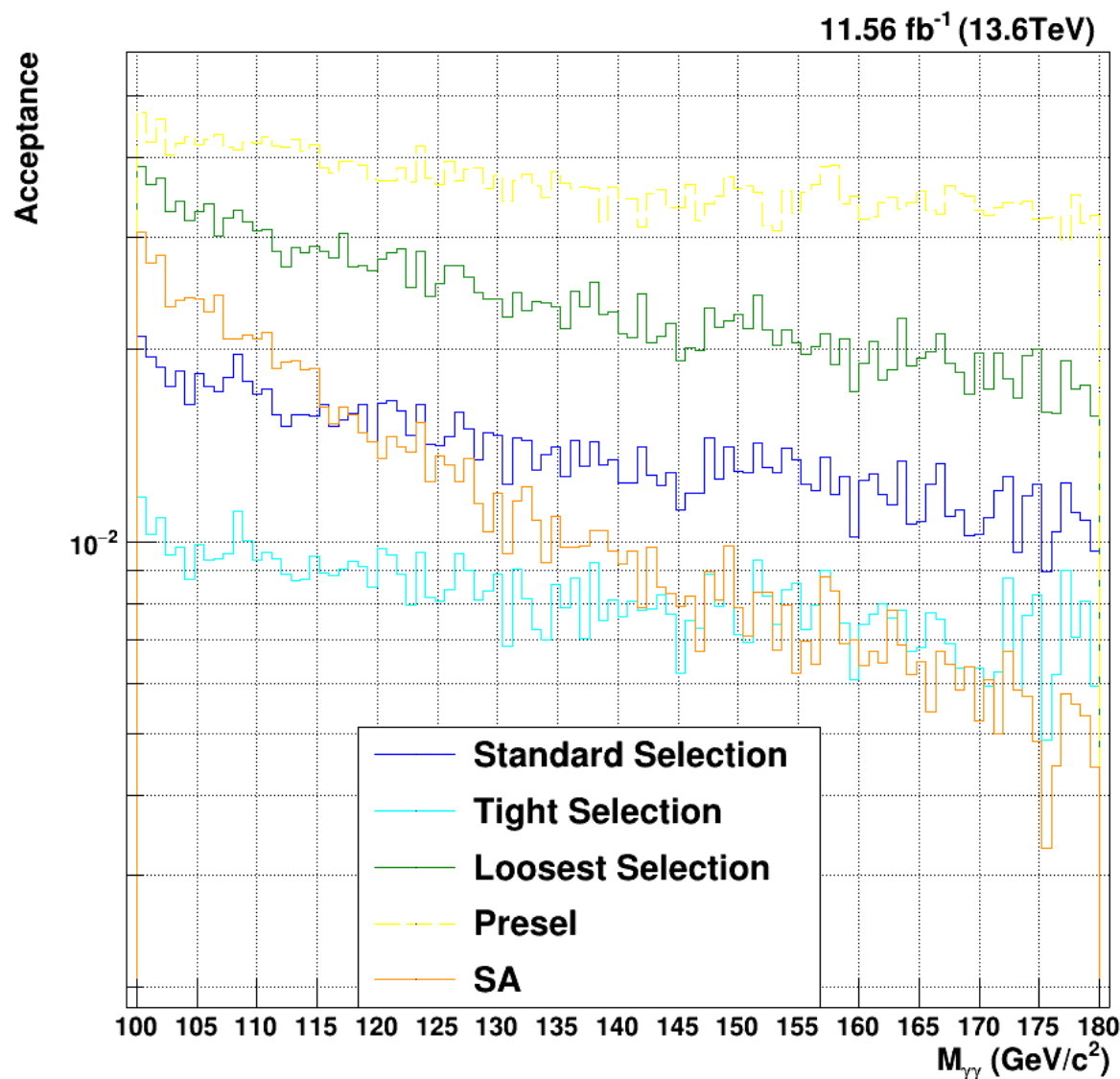


Data



- The signal efficiency and data acceptance can be measured as the percent of the “All” curve accepted by each of the selections.
- The shaping of the background mass distribution by the **Standard Analysis Cuts** is very clear here.
- Meanwhile, the BDT-based selections are relatively flat in background mass.

Data



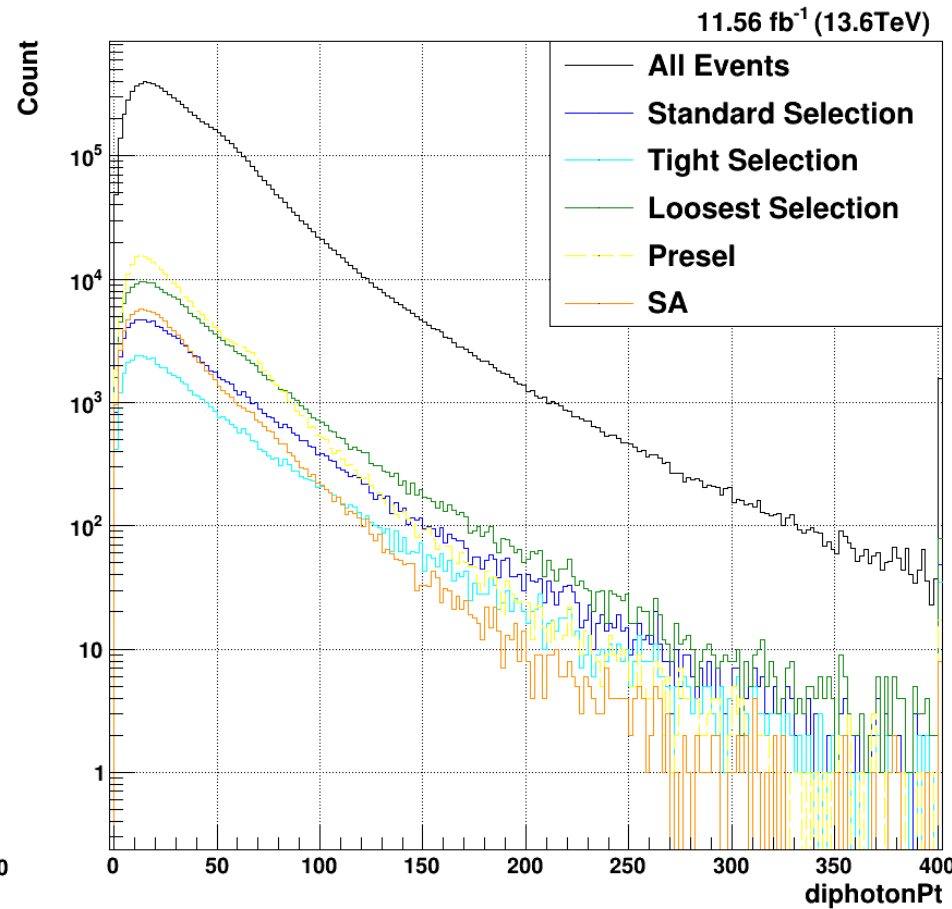
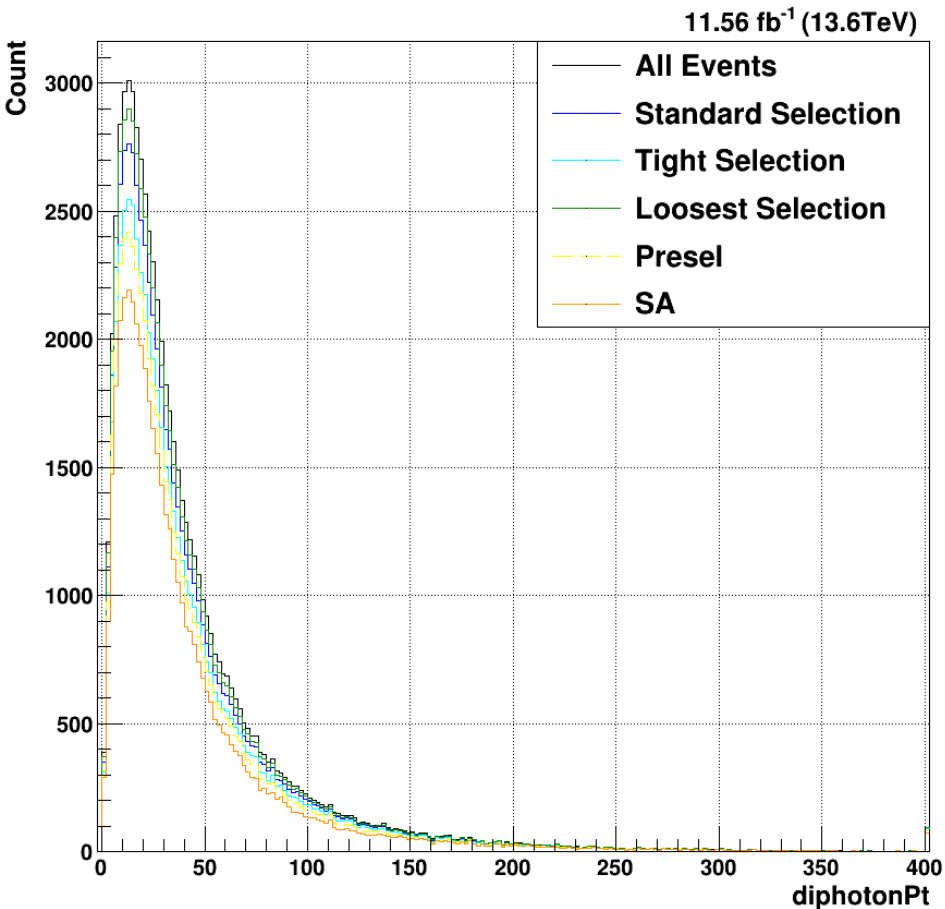
- The signal efficiency and data acceptance can be measured as the percent of the “All” curve accepted by each of the selections.
- The shaping of the background mass distribution by the **Standard Analysis Cuts** is very clear here.
- Meanwhile, the BDT-based selections are relatively flat in background mass.

Single Higgs Diphoton p_T



Higgs MC Signal

Data

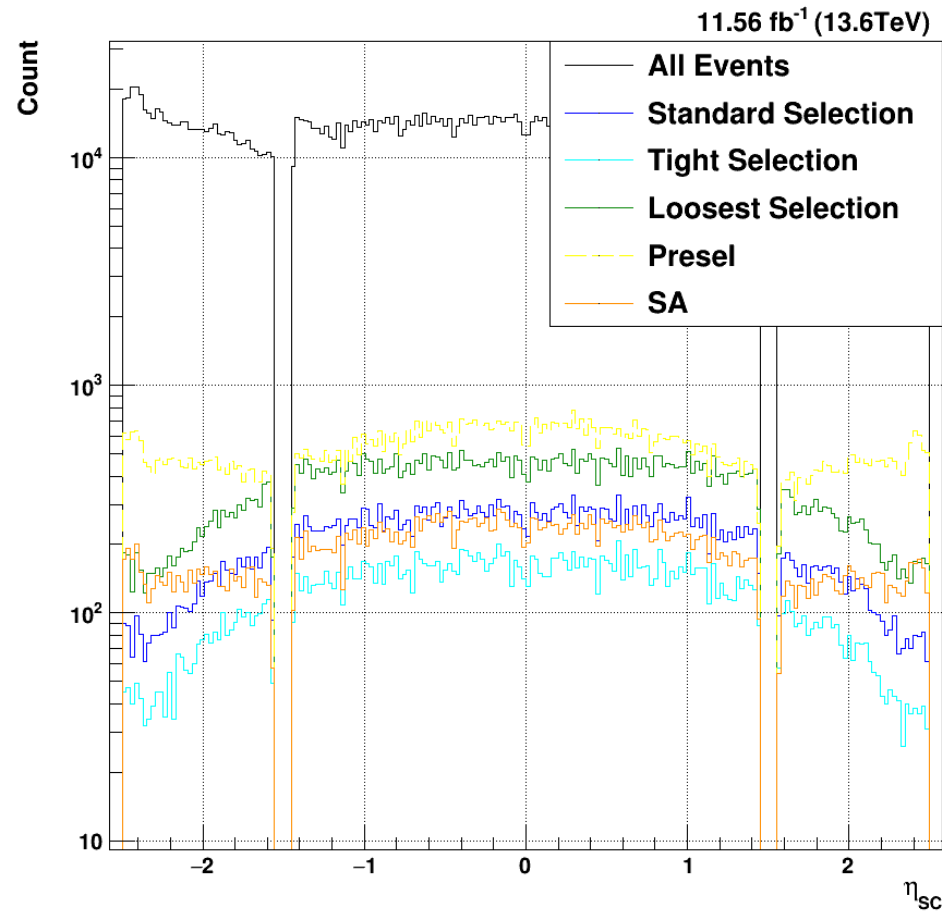
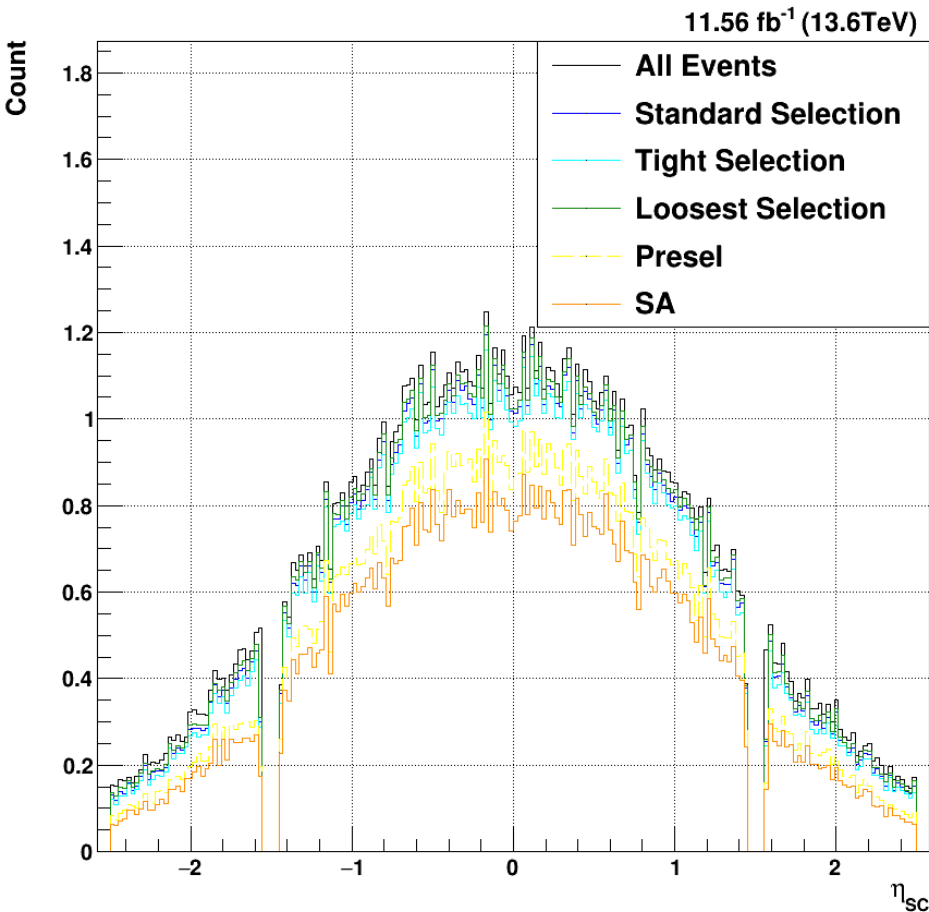


- Neither set of selections noticeably leverages the single Higgs diphoton p_T to make a selection.

- For the single Higgs sample, the diphoton p_T of the signal MC is relatively similar to that of the data.
- This variable will be more interesting for the DiHiggs sample.

DiHiggs MC Signal

Data

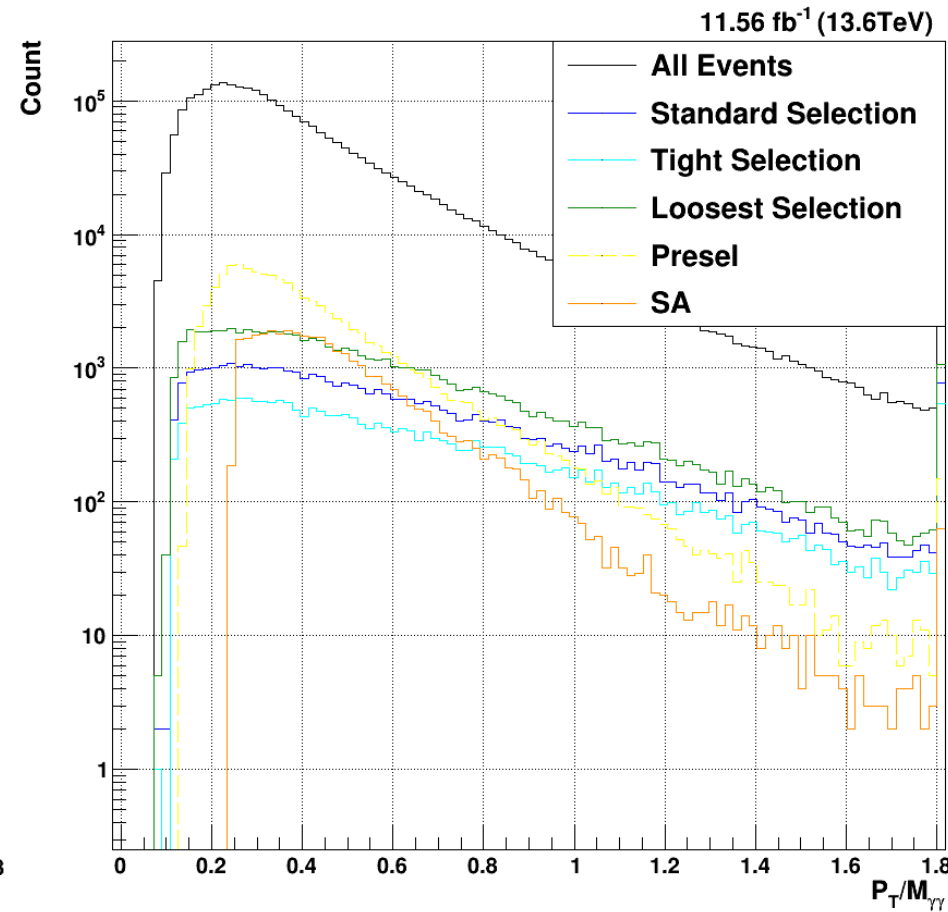
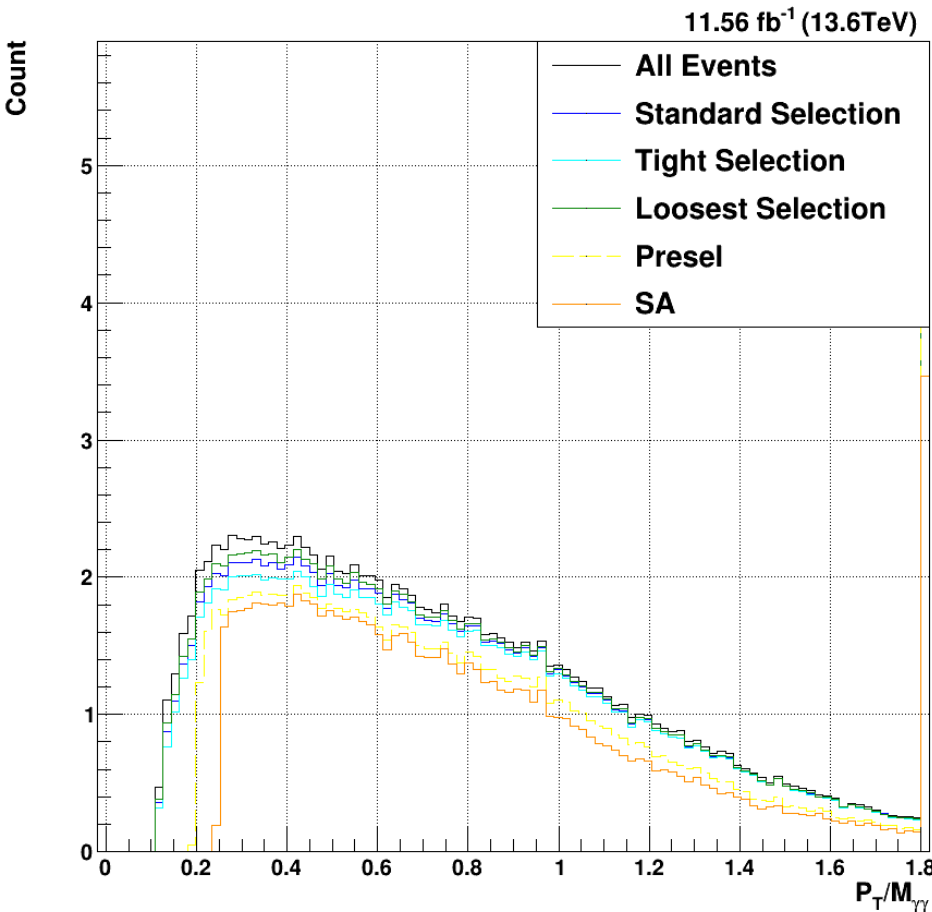


- Again, the kinematic coverage of the BDT-based selection is better than that of the standard analysis selection in η_{SC}

- Similar features can be noted compared to the Single Higgs η_{SC} .
- Note that there are fewer DiHiggs photon candidates in the endcap relative to the single Higgs.

DiHiggs MC Signal

Data



- The $p_T/M_{\gamma\gamma}$ distribution for the DiHiggs signal tends to extend to higher values than that of the single Higgs.
 - Due to the kinematics of having two Higgs produced, which are often back-to-back in the center of mass frame.

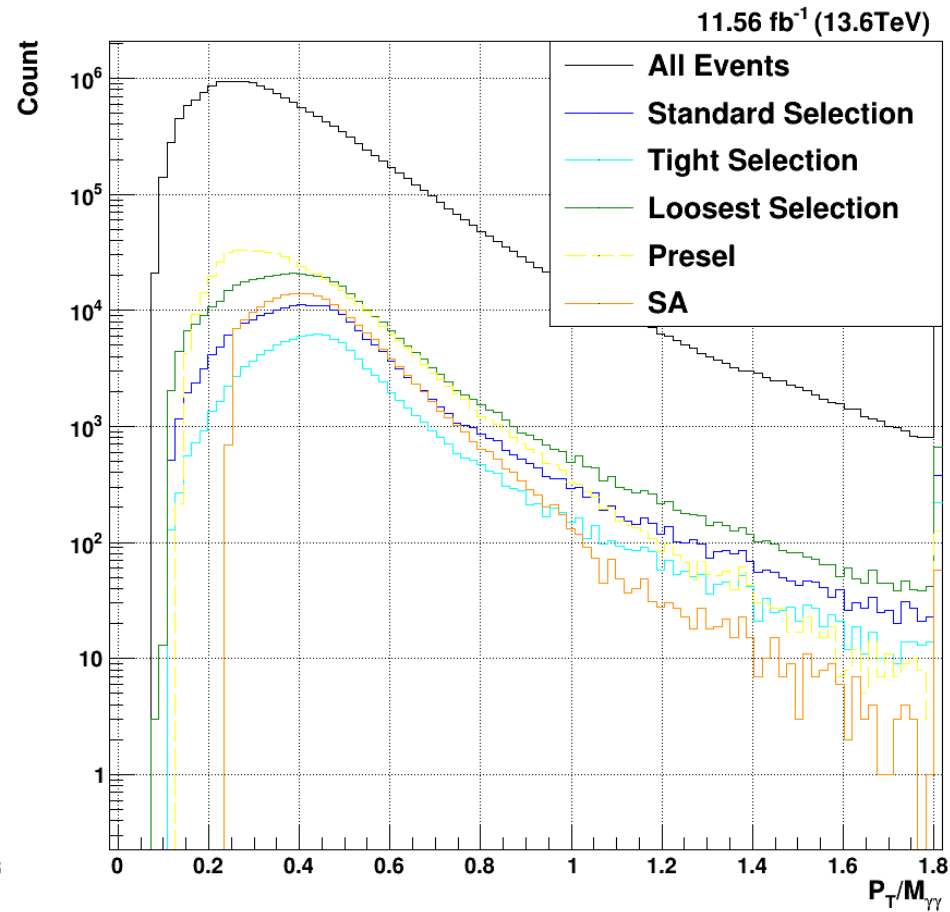
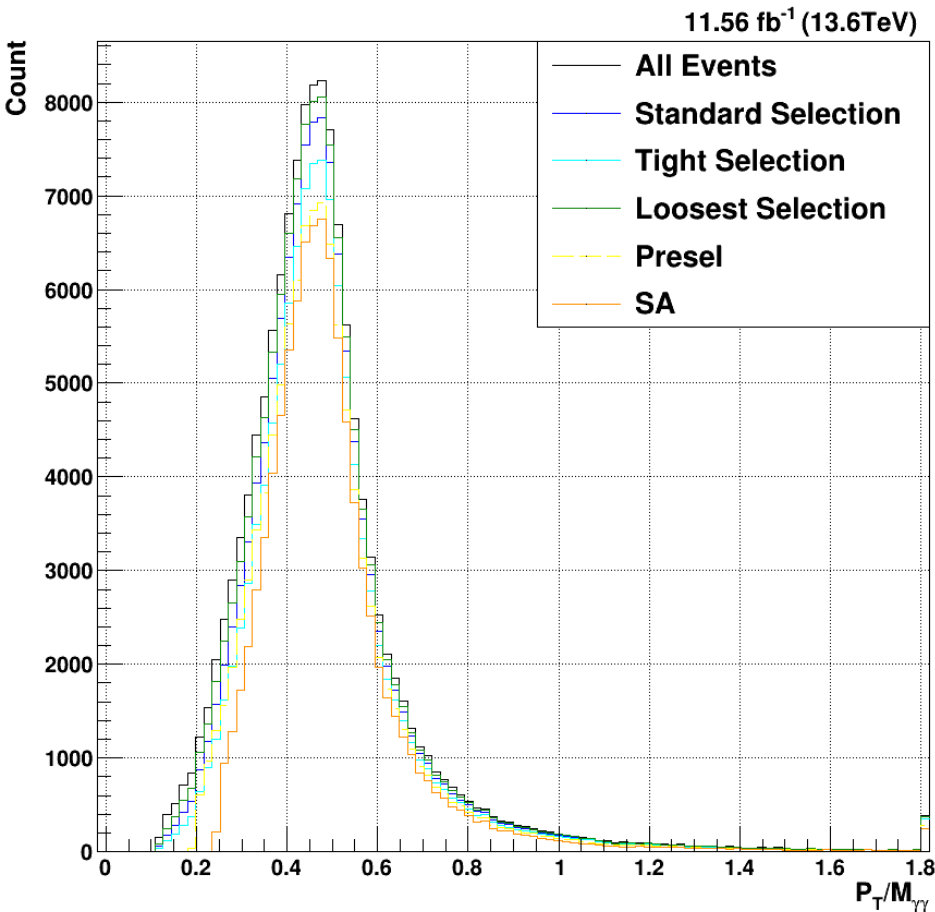
- The hard $p_T/M_{\gamma\gamma}$ cuts still greatly affect the signal efficiency of the SA cuts, so removing them both increases the kinematic coverage and improves the overall signal efficiency considerably.

Single Higgs $p_T/M_{\gamma\gamma}$



Higgs MC Signal

Data



- The SA cuts on $p_T/M_{\gamma\gamma}$ are helpful for controlling the background, but also clearly reduce the signal efficiency considerably.

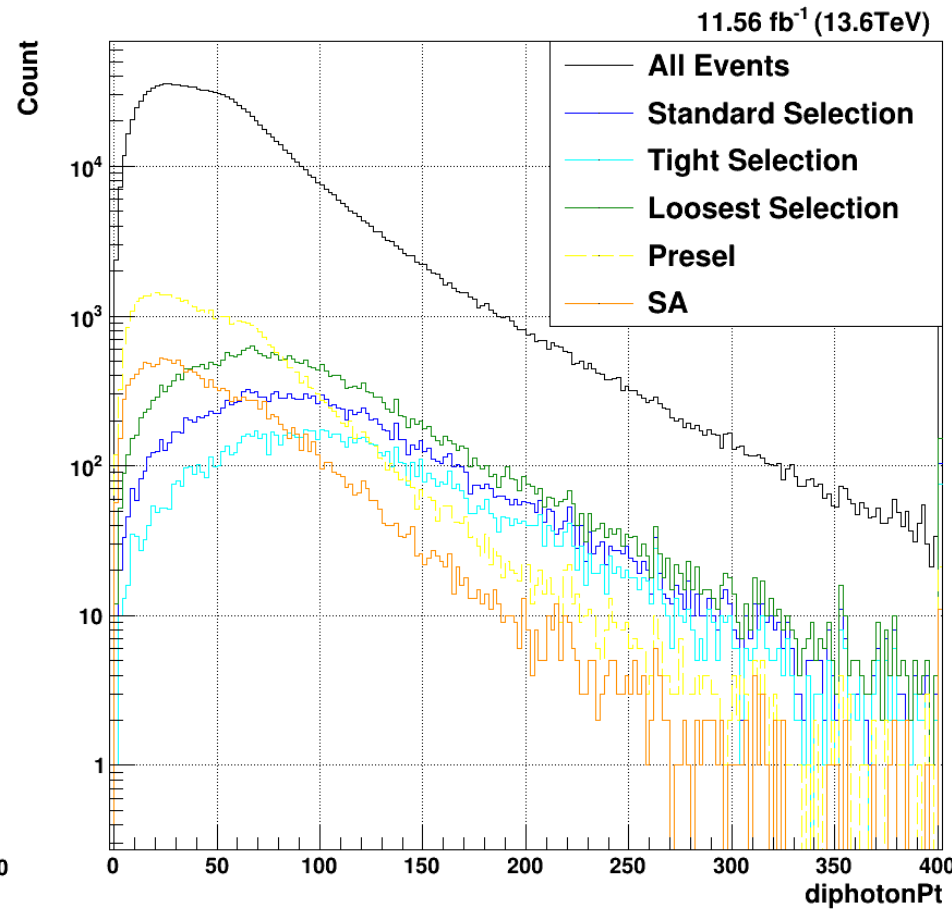
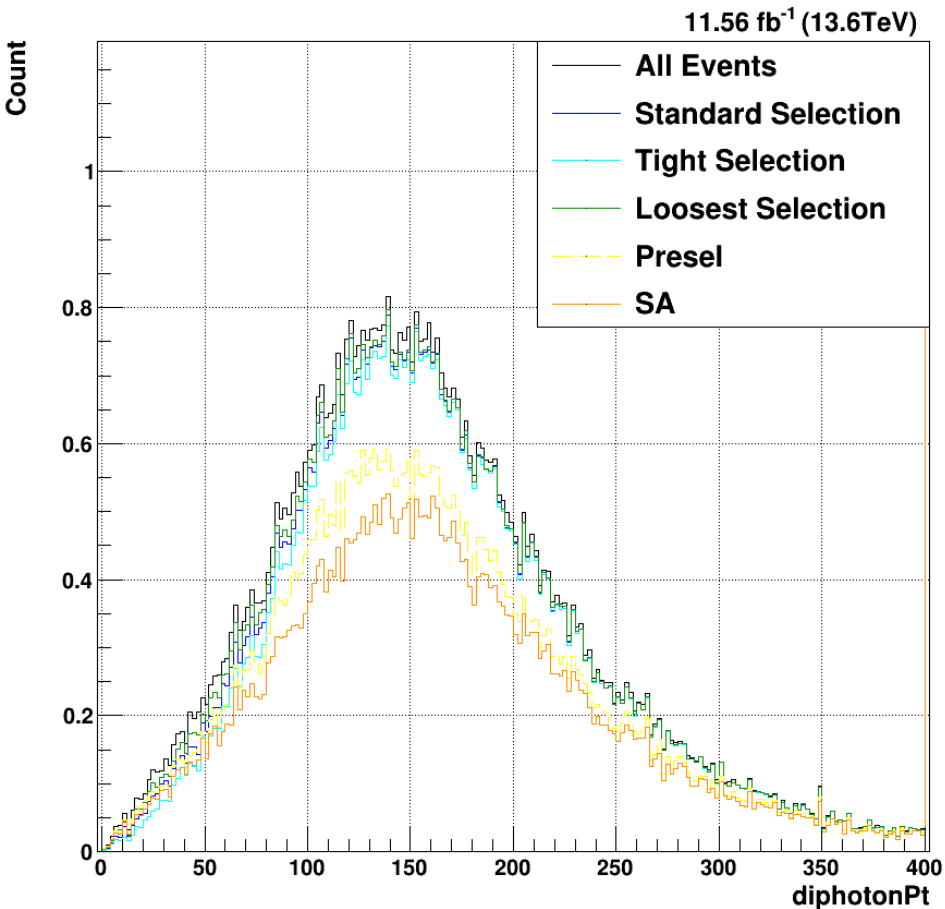
- By using the diphoton BDT scores instead of hard cuts on the $p_T/M_{\gamma\gamma}$, the BDT-based approach is able to further improve the kinematic coverage of the offline selection.

DiHiggs Diphoton p_T



DiHiggs MC Signal

Data



- For the DiHiggs, the Diphoton Pt is used by the diphoton BDT to reject data.

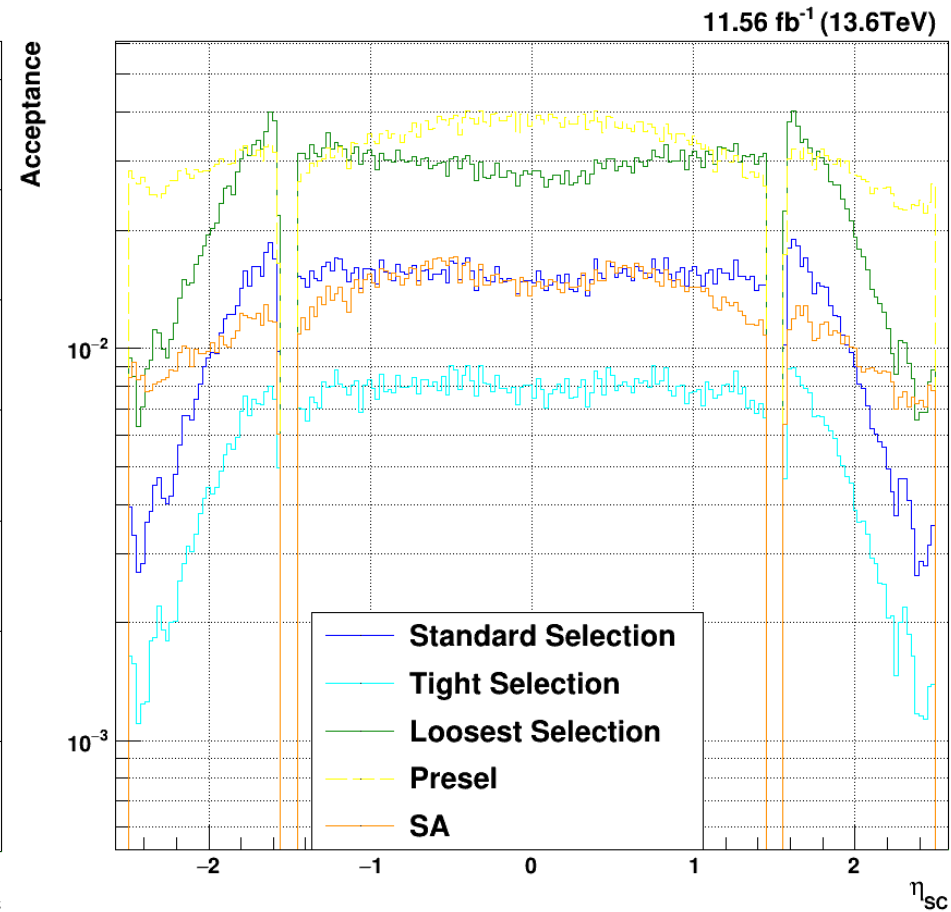
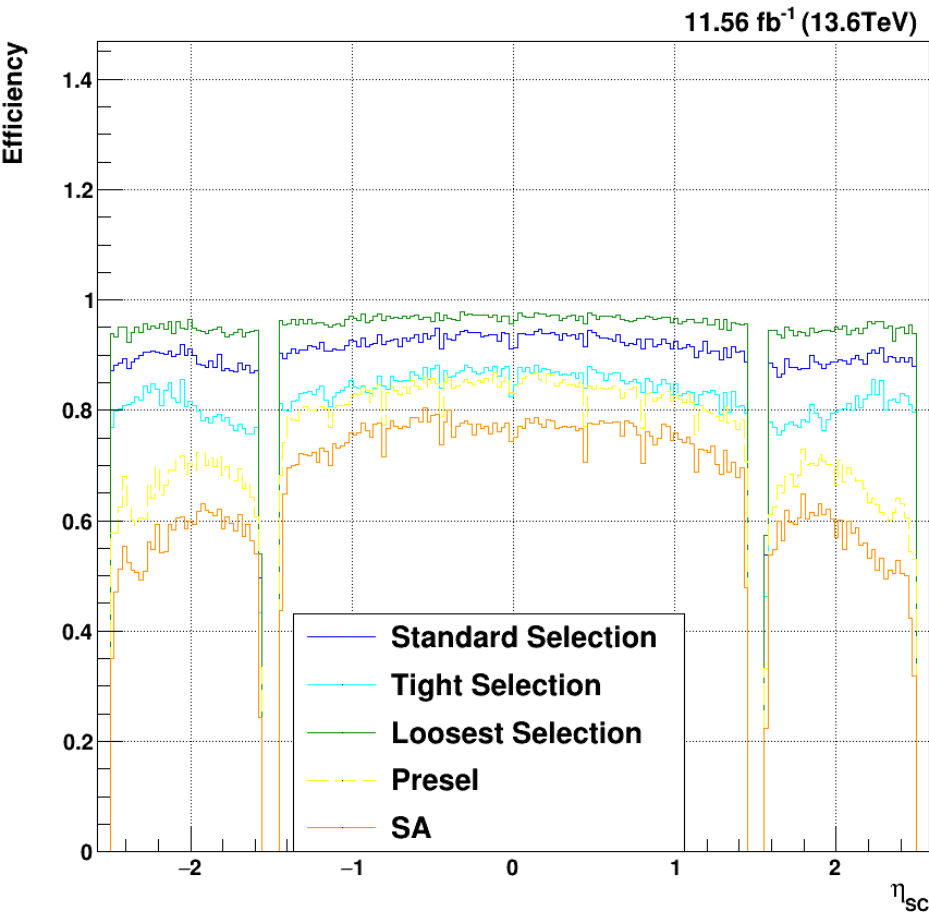
- The Diphoton p_T of the DiHiggs is quite different from that of the Single Higgs, extending to considerably higher values.
 - Since two Higgs are produced back-to-back, each one is more likely to have high p_T .
- In the data, the BDT-based selections are clearly biased towards accepting higher values of Diphoton p_T .

Single Higgs η_{SC}



Higgs MC Signal

Data



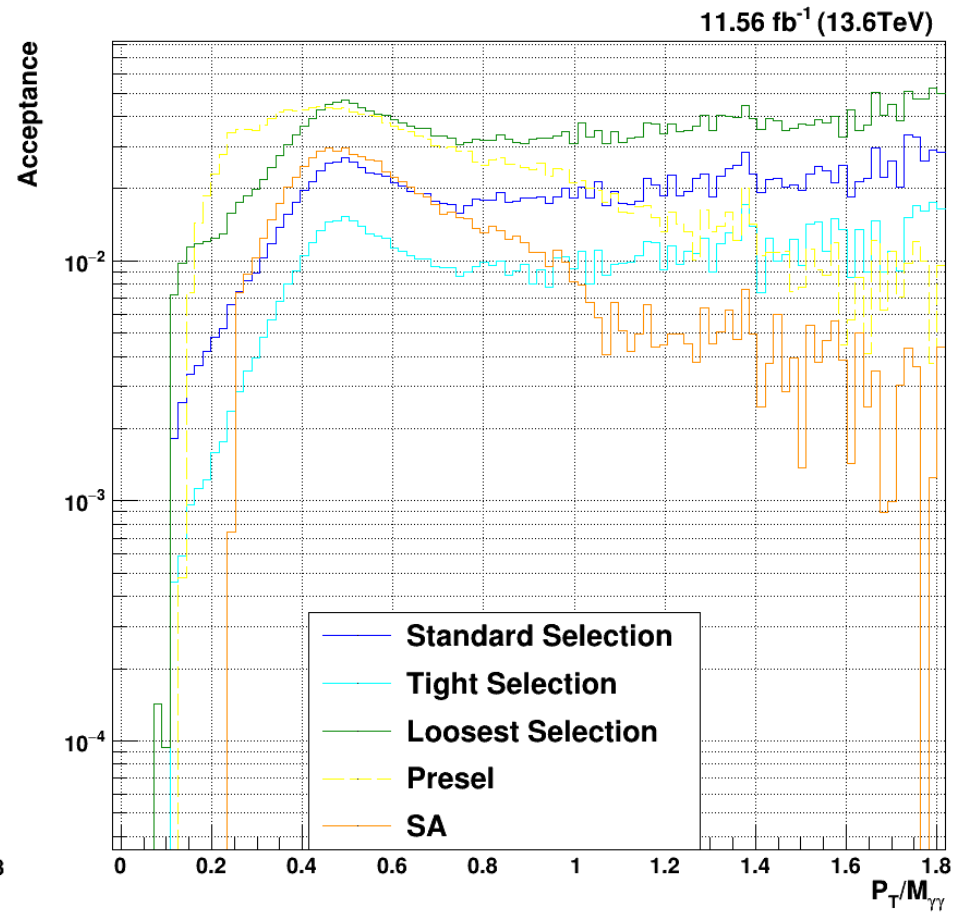
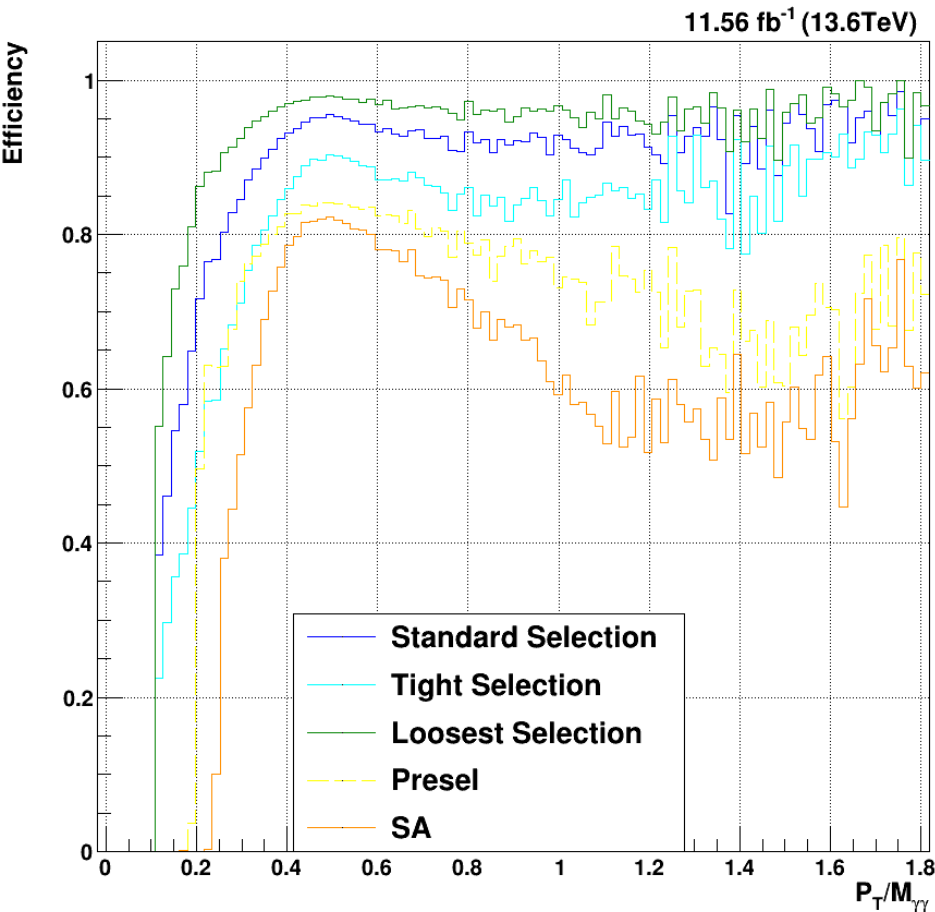
- The inefficiencies of the SA cuts are clearer here.
- The BDT-based selection keeps relatively flat efficiency throughout the selected η_{SC} range.

Single Higgs $p_T/M_{\gamma\gamma}$



Higgs MC Signal

Data



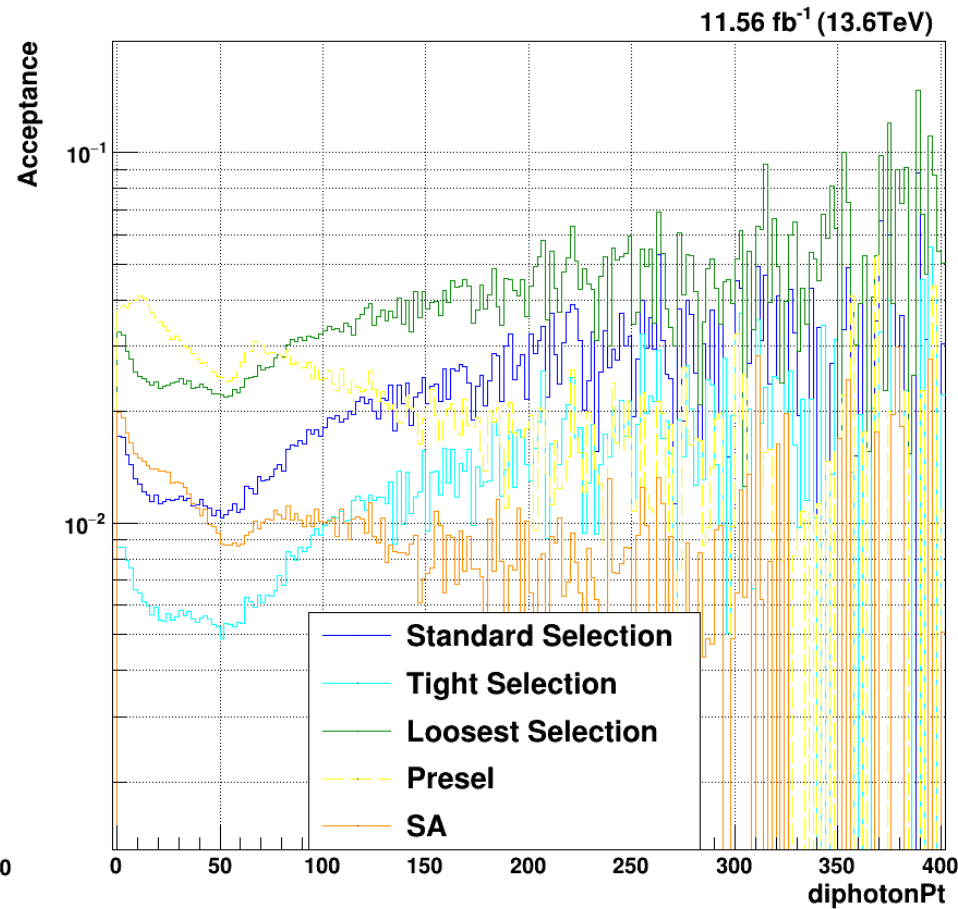
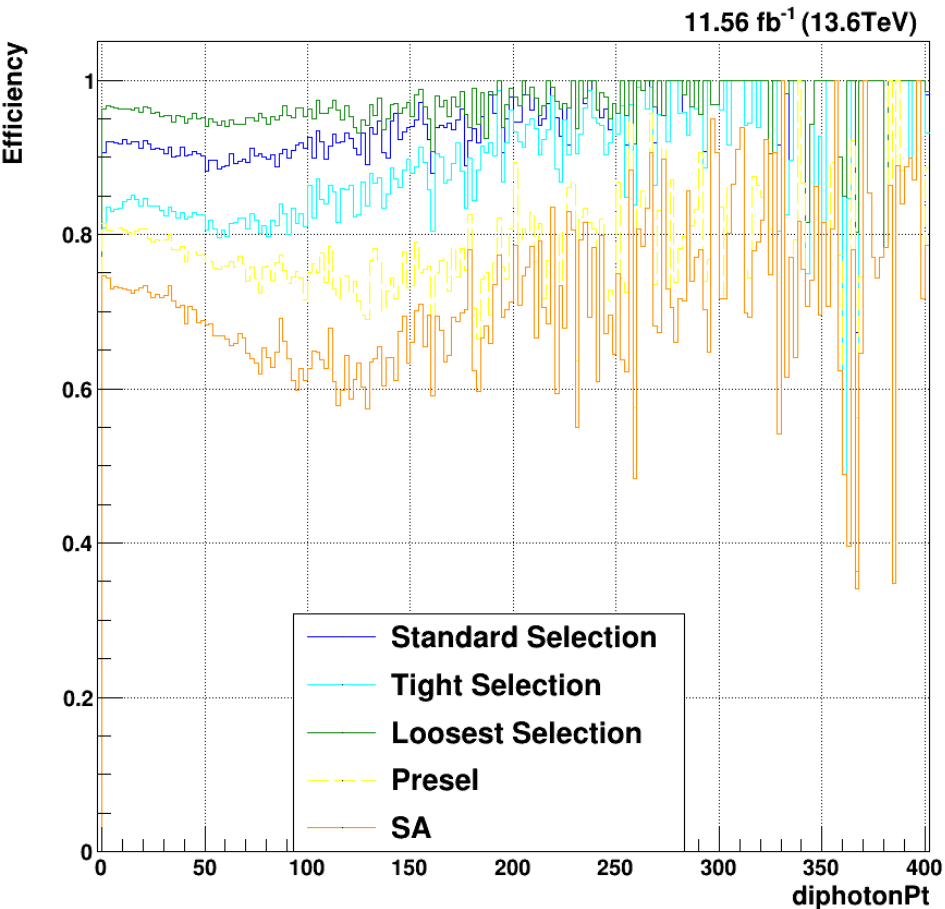
- The Presel and SA cuts both reject not only low- $p_T/M_{\gamma\gamma}$ photons but also reduce the efficiency at higher values.
- The efficiency of the BDT-based approach is relatively flat above $p_T/M_{\gamma\gamma} = 0.4$.

Single Higgs Diphoton p_T



Higgs MC Signal

Data



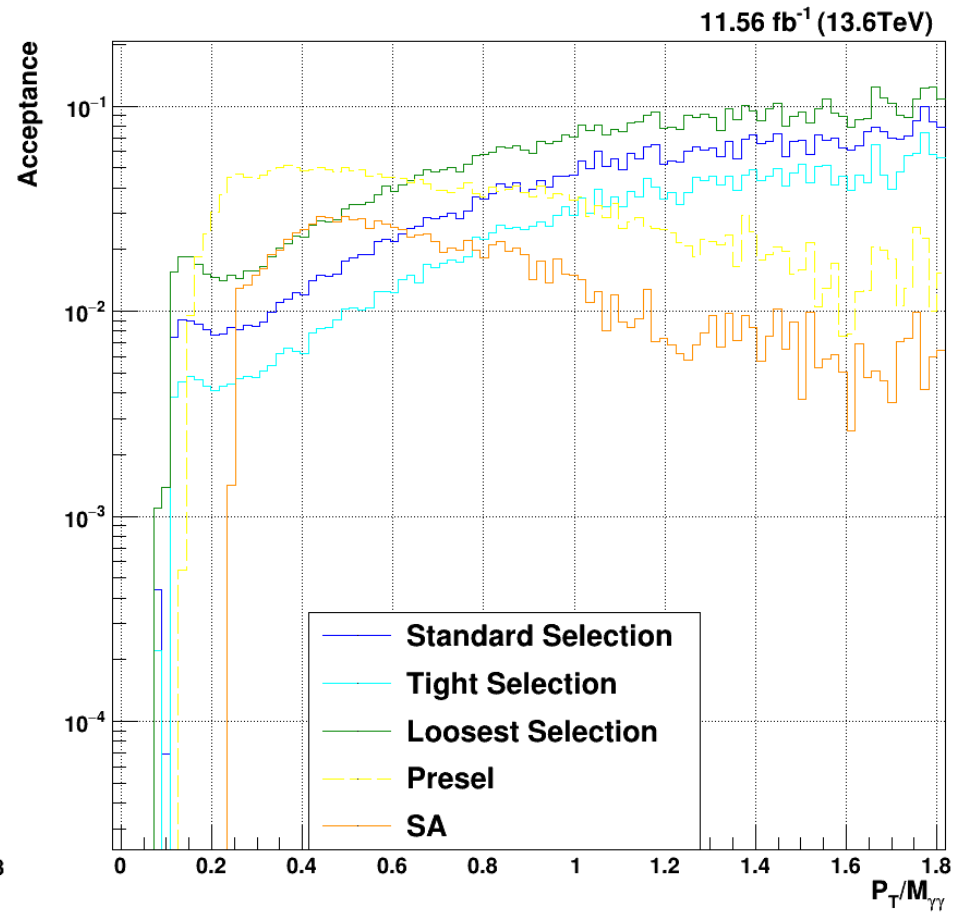
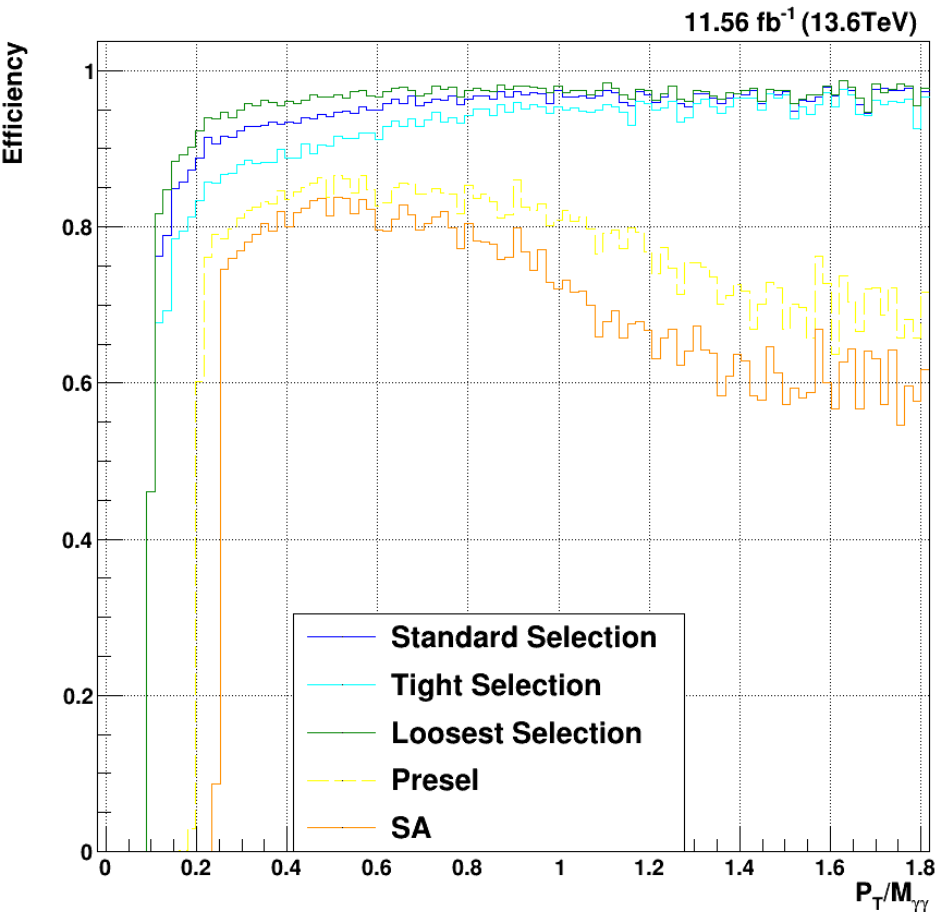
- Again, the BDT-based selection has a flatter efficiency across the whole range of diphoton p_T .

DiHiggs $p_T/M_{\gamma\gamma}$



DiHiggs MC Signal

Data



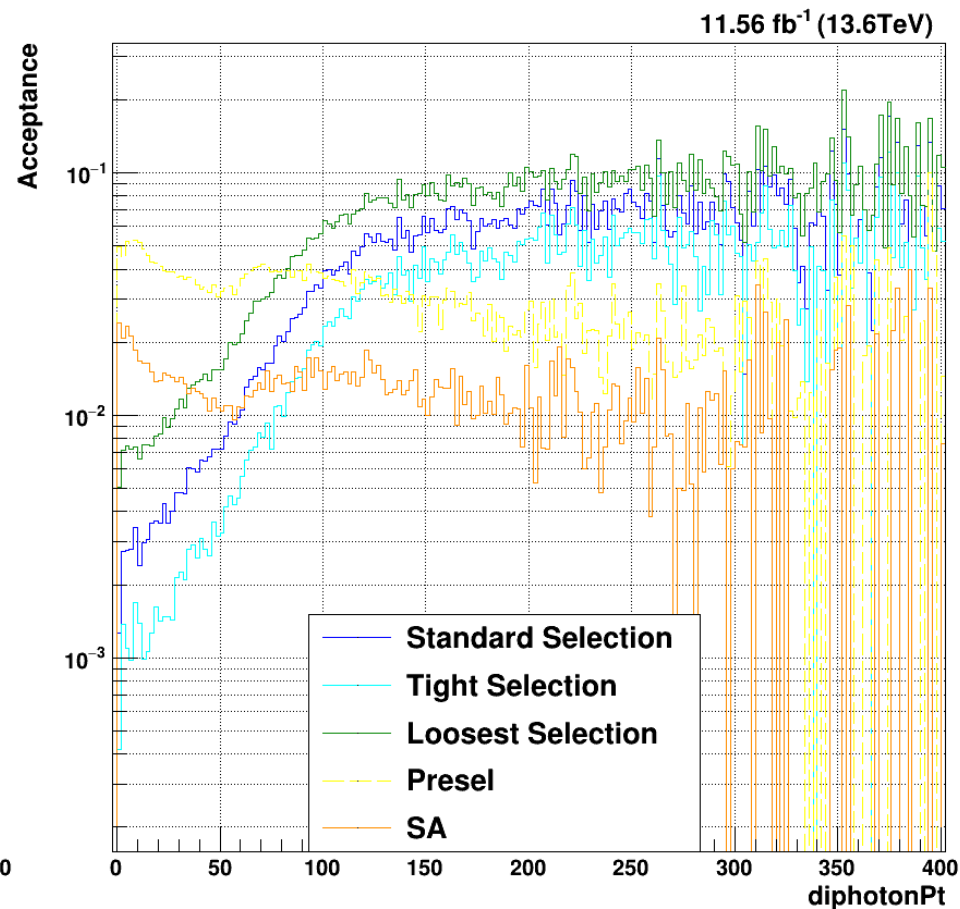
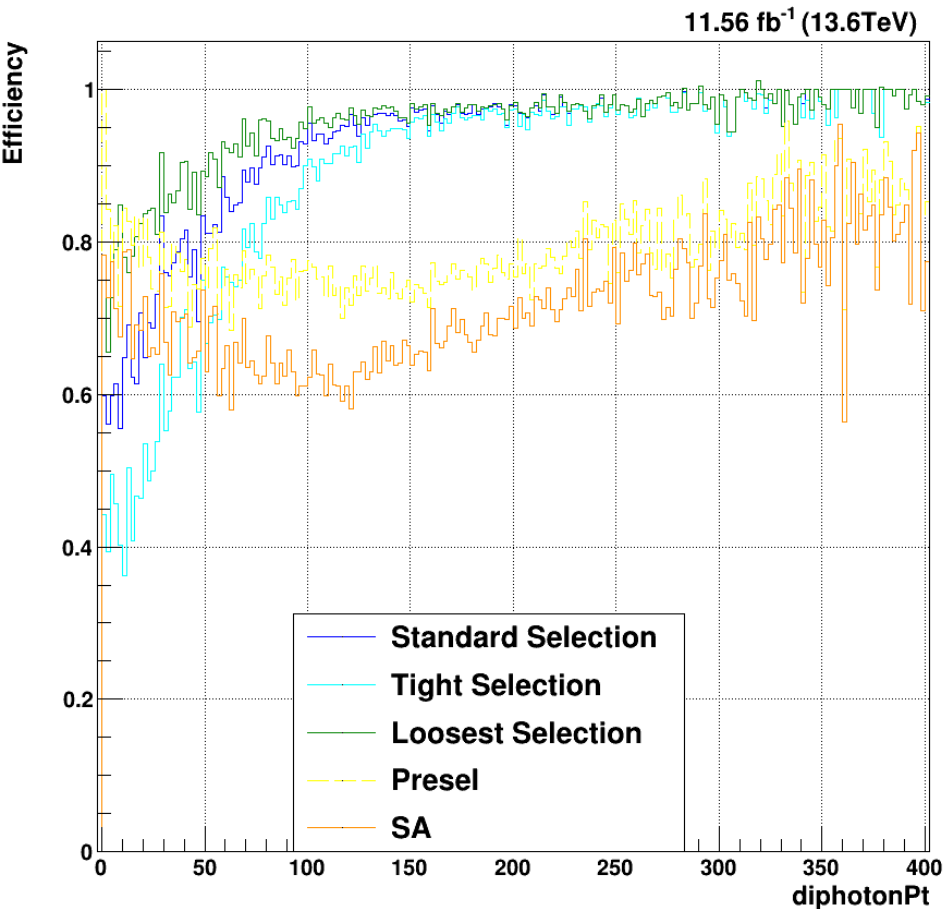
- The Presel and SA cuts both reject not only low- $p_T/M_{\gamma\gamma}$ photons but also reduce the efficiency at higher values.
- The signal efficiency of the BDT-based approach is relatively flat above $p_T/M_{\gamma\gamma} = 0.3$.

DiHiggs Diphoton p_T



DiHiggs MC Signal

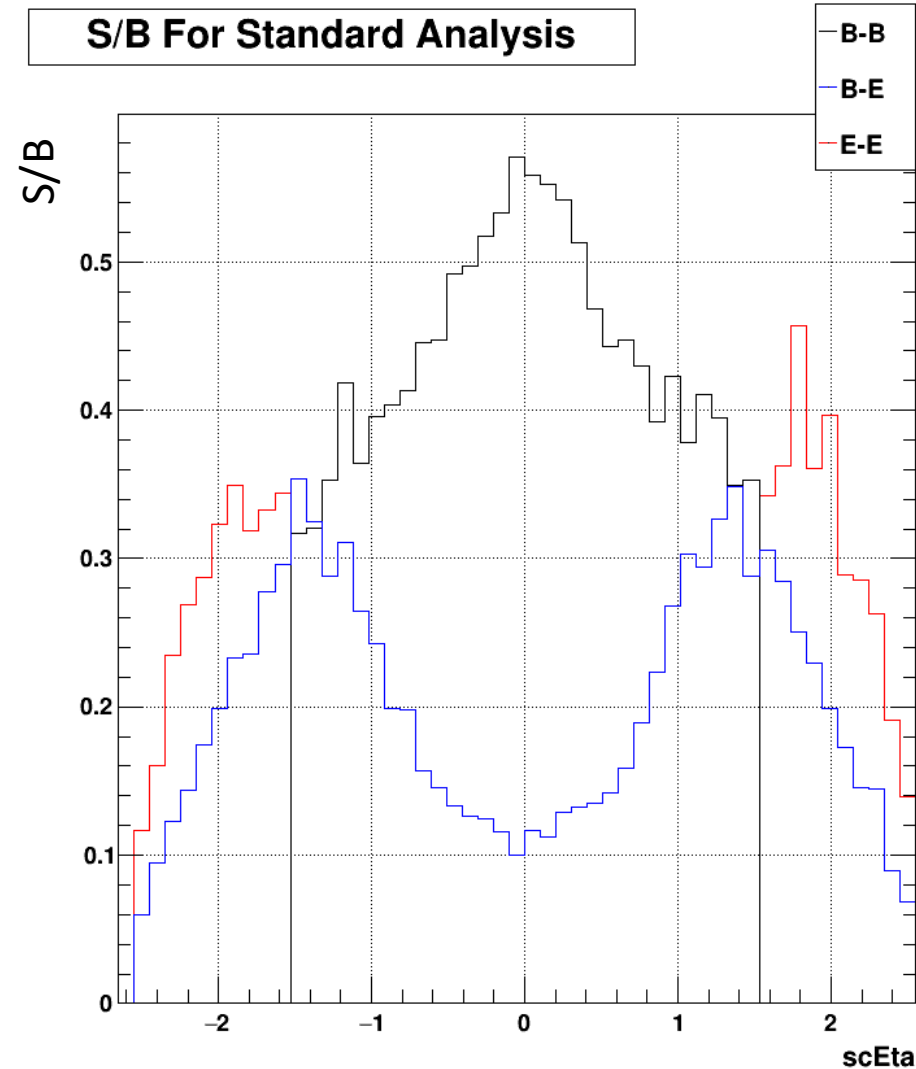
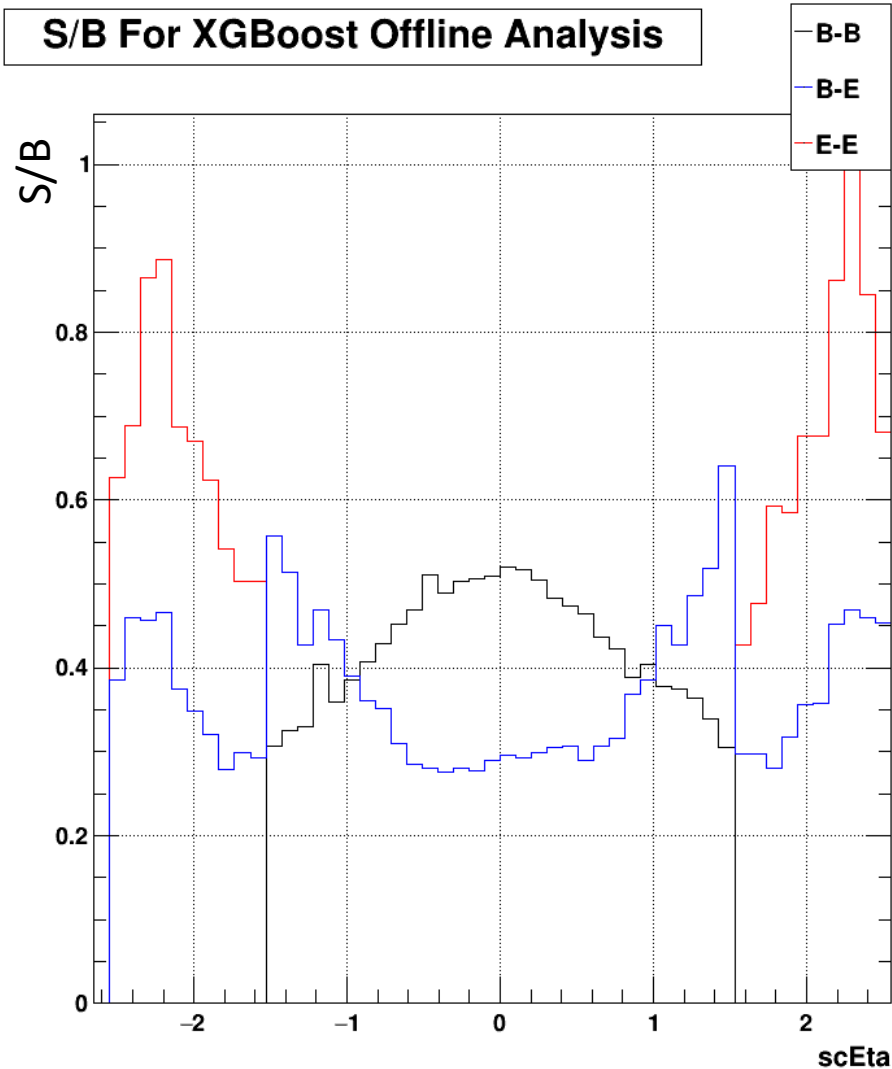
Data



- The diphoton BDT clearly prefers higher diphoton p_T for both the signal and background selection.

- For the signal, beyond the initial inefficiency at lower diphoton p_T (where there aren't many events), the BDT-based selections are quite flat.

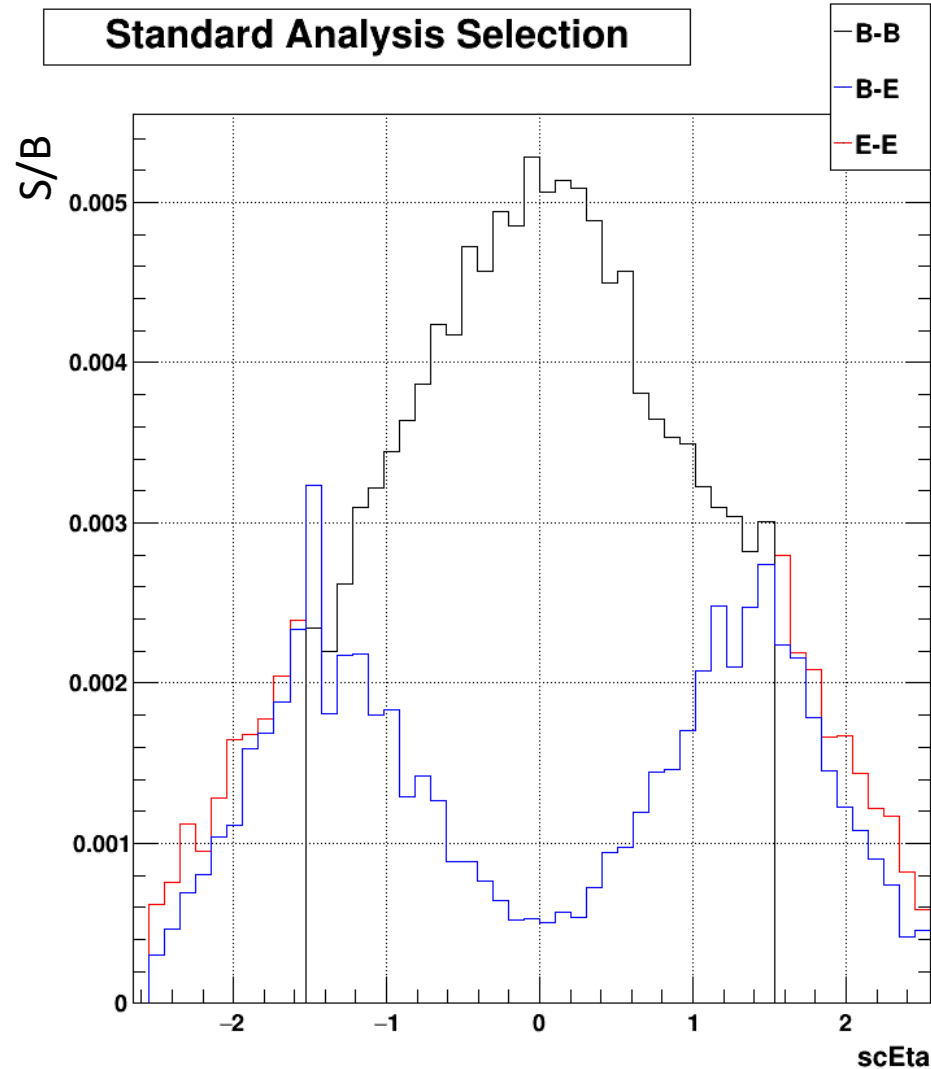
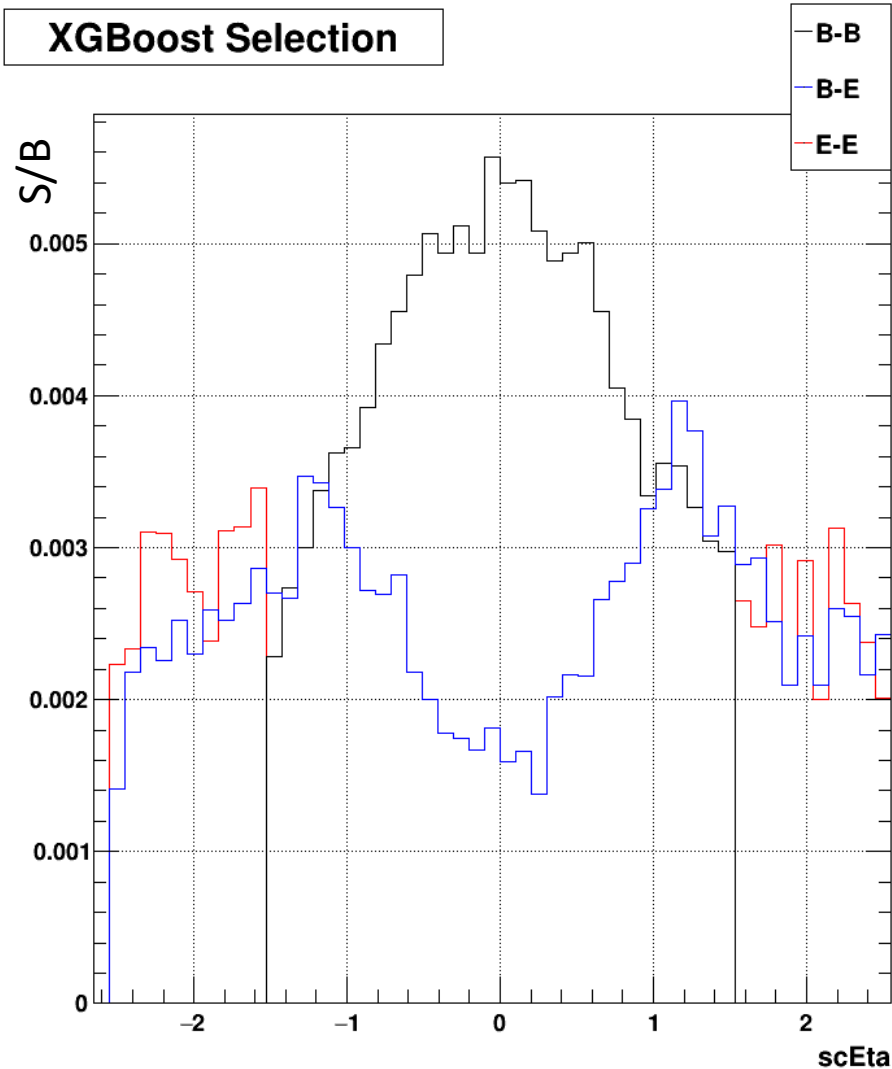
Single Higgs S/B Comparison for η_{SC}



- The (weighted) ratio of the signal to background can be constructed for both the BDT-based selection (Left) and the Standard Analysis (Right)
- Our S/B for the B-B events seems relatively equivalent to that of the Standard Analysis in the barrel.
- For events with one or two photon candidates in the endcap, however, our S/B has increased considerably.
 - We saw that we were cutting out more background with increasing η , while also preserving more signal.

The S/B of the BDT-Based selection in the BE and EE events is much higher than that of the Standard analysis.

DiHiggs S/B Comparison for η_{sc}



- For the DiHiggs, the overall S/B is much lower given that σ_{HH} is about 1000x smaller than σ_H .
- Similar improvements to the S/B in B-E and E-E events can be seen, however.

The S/B of the BDT-Based selection in the BE and EE events is much higher than that of the Standard analysis.